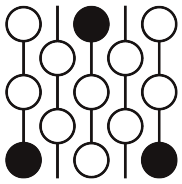
Review of Problem 2 – Humanize



First Proof Third Batch
Reviewer 3
October 8–9, 2026

1 Review

Summary

The approach presented here differs substantially from the other AI-based solutions. The presented approach here reconstructs the unknown conductivity f by a least squares procedure after projecting the data onto a finite-dimensional subspace of L^2. The estimation error is analysed using ideas and techniques from multiresolution analysis. However, the main result (Theorem 3.1), which claims that

$$cn^{-\frac{2\alpha}{2\alpha+d+2}} \leq \inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\left[\|\hat{f}_n - f_0\|^2_{L^2(D)}\right] \leq Cn^{-\frac{2\alpha}{2\alpha+d+2}},$$

is incorrect. The claimed convergence rate is too fast: the denominator should contain $d+3$, rather than $d+2$. This error appears to come from a insufficient treatment of the spatially varying ill-posedness of the underlying inverse problem. The issue can be traced to equation (19), Lemma 7.2, and Remark 7.4. Although the final sentence of Lemma 7.2 highlights the degeneracy of the transport field for modes localised near the origin, the risk analysis does not adequately account for its effect on the estimation error, which is one of the main key points in this problems. Beyond this conceptual issue, the manuscript contains several technical inconsistencies, including in the construction of the wavelet set $\mathcal{F}_\delta$. The exposition is often too unclear to guarantee a detailed assessment of the arguments. Moreover, the manuscript makes no substantive attempt to place its contribution within the existing literature. Taken together, the manuscript does not meet the standards of a scientific publication.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: -1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

3 Problem and answer

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, be the Euclidean unit ball. For a Hölder–Zygmund conductivity $f \in C^\alpha(D)$, $\alpha > 1$, let $u = u_f$ solve the Darcy problem

$$\operatorname{div}(f\nabla u) = 1 \ \text{ on } D, \qquad u|_{\partial D} = 0.$$

We observe

$$Y \ = \ u_f \ + \ \frac{1}{\sqrt{n}}\,\mathbb{W},$$

where $\mathbb{W}$ is Gaussian white noise on $L^2(D)$, and we estimate f over the class

$$\mathcal{F}(\alpha, r) = \{\, h \in C^\alpha(D) : \ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r \,\}, \qquad r \in (0,1) \text{ small and fixed.}$$

(The class is contained in C^a with the a-th derivatives uniformly t-Hölder, where $\alpha = a+t$, $a = \lceil \alpha \rceil - 1 \in \mathbb{Z}_{\geq 1}$, $t \in (0,1]$, Zygmund if $t = 1$; in particular every $f \in \mathcal{F}(\alpha, r)$ is continuously differentiable and satisfies $1 - r \leq f \leq 1 + r$.)

Theorem 3.1. *There exist $r_0 = r_0(\alpha, d) \in (0,1]$ and, for every fixed $r \in (0, r_0]$, constants $c, C > 0$ with C depending only on (α, d) and c on (α, d, r), such that*

$$c\,\Phi_n \ \leq \ \inf_{\widehat{f}_n} \ \sup_{f_0 \in \mathcal{F}(\alpha, r)} \mathbb{E}_{f_0}\big[\|\widehat{f}_n - f_0\|^2_{L^2(D)}\big] \ \leq \ C\,\Phi_n, \qquad \Phi_n = n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

Answer. The minimax rate is

$$\boxed{n^{-\frac{2\alpha}{2\alpha+d+2}}} \qquad \text{up to logarithmic factors.}$$

Both bounds proved below carry this power of n; our argument is in fact log-free, which is a fortiori "up to logarithmic factors." This is *faster* than the direct white-noise rate $n^{-2\alpha/(2\alpha+d)}$: the denominator gains an additive $+2$, i.e. the inverse problem is only *mildly ill-posed of degree one*. The mechanism, proved below, is that the linearization T of $f \mapsto u_f$ at $\mathbf{1}$ satisfies $\Delta(Tg) = g + \frac{x\cdot\nabla}{d} g$, and the transport-type source $g \mapsto g + \frac{x\cdot\nabla}{d} g$ is invertible as an order-zero operator ($L^{-1} = J \in \mathcal{B}(L^2(D))$ with $\|J\| \leq 2d$, $L \in \mathcal{B}(H^1(D), L^2(D))$; Lemma 4.3); consequently T behaves like a general (nonlocal) operator of order -1, gaining exactly one derivative on scale-δ perturbations, rather than two.
The lower bound is Proposition 6.1; the upper bound is Proposition 7.1. Constants implicit in $\lesssim, \asymp$ depend only on (α, d) and a fixed bump, never on f, n, or δ; C is uniform in $r \in (0, r_0]$, while the lower-bound c carries the factor r^2 of the shrinking class (Proposition 6.1).

4 Preliminaries

4.1 White-noise model and likelihood

Gaussian white noise $\mathbb{W}$ on $L^2(D)$ is the centred Gaussian process indexed by $L^2(D)$ with $\mathbb{E}\langle \mathbb{W}, \phi\rangle = 0$ and $\mathbb{E}[\langle \mathbb{W}, \phi\rangle\langle \mathbb{W}, \psi\rangle] = \langle \phi, \psi\rangle_{L^2(D)}$. Under conductivity f, the law P_f of Y is the Gaussian shift with mean u_f and covariance $n^{-1}I$ on $L^2(D)$. For two conductivities f, f_0 the Cameron–Martin formula gives

$$\mathrm{KL}(P_f \,\|\, P_{f_0}) \ = \ \frac{n}{2}\,\|u_f - u_{f_0}\|^2_{L^2(D)}. \tag{1}$$

All norms $\|\cdot\|$ are $L^2(D)$ unless otherwise indicated; $\|\cdot\|_p$ denotes $L^p(D)$.

4.2 External tools (precise statements)

We collect the standard results used, with precise hypotheses, so that no nonstandard result enters tacitly.

> **Reviewer 3**
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> Here, standard results are stated without appropriate references.

(T1) **Fano's inequality.** For probability measures $P_1, \dots, P_N$ on a common space, any estimator $\widehat{\theta}$ of the index, and any reference $\bar{P}$, $\frac{1}{N}\sum_j P_j(\widehat{\theta} \neq j) \geq 1 - \frac{\frac{1}{N}\sum_j \mathrm{KL}(P_j \| \bar{P}) + \log 2}{\log N}$.

(T2) **Varshamov–Gilbert.** For $M \geq 1$ there exist $N \geq 2^{cM}$ subsets $S \subset \{1, \dots, M\}$, each of cardinality $M/2$, with pairwise Hamming distance $|S \triangle S'| \geq M/8$; $c > 0$ absolute.

(T3) **Weyl's law.** For the Dirichlet Laplacian on the bounded smooth domain D, $\#\{\lambda_\nu \leq \Lambda\} \sim c_d |D| \Lambda^{d/2}$; hence the span of the eigenfunctions with $\lambda_\nu \leq M^2$ has dimension $\asymp M^d$.

(T4) **Cameron–Martin.** For $Y = u + n^{-1/2}\mathbb{W}$ on $L^2(D)$, means $u_1, u_0 \in L^2(D)$ give laws with $\mathrm{KL}(P_{u_1} \| P_{u_0}) = \frac{n}{2}\|u_1 - u_0\|_2^2$.

(T5) **Wavelet/Besov characterization of C^α and square function on D.** There is a boundary-corrected wavelet frame of the smooth domain D (Cohen–Dahmen–DeVore-type) such that for $\alpha > 0$ (noninteger, with the Zygmund convention at integers), $f \in C^\alpha(D) = B^\alpha_{\infty,\infty}(D)$ iff $\sup_\lambda 2^{j_\lambda(\alpha + d/2)} |\langle f, \psi_\lambda \rangle| < \infty$, comparable to $\|f\|_{C^\alpha}$; and $\|f\|_{L^2} \asymp \big(\sum_j \|\Delta_j f\|_{L^2}^2\big)^{1/2}$ for the spectral dyadic pieces $\Delta_j = \varphi_j(\sqrt{-\Delta})$ (the Dirichlet-Laplacian smooth functional calculus / heat-kernel Mikhlin square-function inequality on the smooth bounded domain).

(T6) **Divergence-form elliptic regularity.** For $1 - r \leq f \leq 1 + r$, $f \in C^\alpha$, $\alpha > 1$, $\mathrm{div}(f \nabla u) = F$ in D with $u = 0$ on ∂D has a unique solution with $\|u\|_{C^{1,\beta}} \leq C(\alpha, d, r)\|F\|_{L^\infty}$ for some $\beta = \beta(\alpha, d) \in (0, 1]$; in particular $\|\nabla u\|_{L^\infty} \leq C$.

(T7) **Hardy / Cesàro.** The Cesàro operator $(Cf)(t) = \frac{1}{t}\int_0^t f(s)\, ds$ has norm $\frac{p}{p-1}$ ($= 2$ on L^2) on $L^p((0,1), dt)$.

Every PDE/functional bound below is either derived here or is an application of (T1)–(T7).

4.3 Linearization about the constant conductivity

The constant-conductivity solution is explicit:

$$u_0(x) = \frac{|x|^2 - 1}{2d}, \qquad \nabla u_0(x) = \frac{x}{d}, \qquad \Delta u_0 = 1.$$

For any $\phi \in C^1(D)$ one has the pointwise identity

$$\mathrm{div}(\phi \nabla u_0) = \nabla\phi \cdot \Big(\frac{x}{d}\Big) + \phi\, \Delta u_0 = \frac{x \cdot \nabla \phi}{d} + \phi. \tag{2}$$

Lemma 4.1 (Linearization)**.** *For $g \in L^2(D)$, let $Tg \in H_0^1(D)$ be the unique solution of $-\mathrm{div}(\nabla w) = -\mathrm{div}(g \nabla u_0)$, $w|_{\partial D} = 0$, i.e. $Tg = -\Delta^{-1}\big(\mathrm{div}(g \nabla u_0)\big)$ in the $H^{-1}(D)/H_0^1(D)$ sense. Then, as distributions,*

$$\Delta(Tg) = g + \frac{x \cdot \nabla}{d}\, g. \tag{3}$$

Proof. By (2), $\mathrm{div}(g \nabla u_0) = g + \frac{x \cdot \nabla}{d} g$ as a distribution of order ≤ 1, hence in $H^{-1}(D)$; applying Δ^{-1} gives (3). The Dirichlet Laplacian $-\Delta : H_0^1(D) \to H^{-1}(D)$ is an isomorphism (Lax–Milgram), and $\|Tg\|_{H^1} \lesssim \|g\|_{H^{-1}}$. □

Remark 4.2 (Transport structure of L). Write $A = \frac{x\cdot\nabla}{d}$ and $L = I + A$. Using $\operatorname{div}(xg) = (\operatorname{div} x)g + x \cdot \nabla g = dg + x \cdot \nabla g$ and $x \cdot \nu = |x| = 1$ on ∂D,

$$Lg = g + \frac{x \cdot \nabla g}{d} = \frac{1}{d}\big(dg + x \cdot \nabla g\big) = \frac{1}{d}\operatorname{div}(x\, g).$$

Hence L is a pure divergence composed with the vector field $x \mapsto x/d$; in particular *if* $\int_D g\,dx = 0$ *then* $\int_D Lg\,dx = \frac{1}{d}\int_{\partial D} g\,dS$ (this vanishes for the mean-zero wavelet packets used below since each ψ_m has integral zero and is interior). This divergence/transport structure makes L an order-zero invertible operator on $L^2(D)$ (Lemma 4.3); the smoothing order of $T = -\Delta^{-1}L$ is then governed by the dilation generator, giving exactly one degree (Lemmas 5.1, 5.3).

4.4 The transport operator L is an invertible order-zero operator

> **Reviewer 3**
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> It is difficult, or even impossible, to verify the correctness of Lemmas 4.3 and 4.5 due to the unclear presentation, lack of justification and missing references.

Lemma 4.3 (L is invertible, $\|L^{-1}\| \le C_d$)**.** *Let* $A = \frac{x\cdot\nabla}{d}$ *and* $L = I + A$. *For* $g \in C^1(D)$ *define*

$$(Jg)(x) \;:=\; d\int_0^1 \tau^{d-1} g(\tau x)\,d\tau, \qquad x \in D \setminus \{0\}.$$

Then:

(i) $LJg = JLg = g$ *on* $C^1(D)$, *so* L *is invertible and* $L^{-1} = J$.

(ii) $\|L^{-1}g\|_{L^2(D)} \le C_d\|g\|_{L^2(D)}$ *for every* $d \ge 2$ *(in fact* $C_d = 2d$ *suffices).*

(iii) Consequently $\|Lg\|_2 \ge c_d\|g\|_2$ *(*$c_d = C_d^{-1} > 0$*). Precisely:* $J = L^{-1}$ *extends to an element of* $\mathcal{B}(L^2(D))$, $L \in \mathcal{B}(H^1(D), L^2(D))$ *(since* $\|Lg\|_2 \le \|g\|_2 + \frac{1}{d}\|\nabla g\|_2$*). Moreover, writing* $Lg = \frac{1}{d}\operatorname{div}(xg)$ *and defining* $\operatorname{div}(xg) \in H^{-1}(D)$ *for general* $g \in L^2(D)$ *by duality, the operator* $T(g) := -\Delta^{-1}\big(\frac{1}{d}\operatorname{div}(xg)\big)$ *extends the* $\Delta w = Lg$ *map of Lemma 4.1 from* C^1 *to all of* $L^2(D)$, *with*

$$T \in \mathcal{B}\big(L^2(D), H_0^1(D)\big) \cap \mathcal{B}\big(H^{-1}(D), L^2(D)\big), \qquad \|T\|_{L^2\to L^2} \le C_d \cdot 2C_P,$$

in particular $\|Tg\|_2 \le C_T\|g\|_2$ *with* $C_T = C_T(d)$, *used in the spectral-tail step of Lemma 7.2.*

Proof. (i) Fix $x \neq 0$ and set $\phi(\tau) = g(\tau x)$, so $\phi'(\tau) = x \cdot \nabla g(\tau x)$. Then

$$\frac{d}{d\tau}\big[\tau^d g(\tau x)\big] = d\tau^{d-1} g(\tau x) + \tau^d x \cdot \nabla g(\tau x).$$

Integrating over $\tau \in [0,1]$ ($\tau^d g(\tau x) \to 0$ as $\tau \to 0$):

$$g(x) = \int_0^1 \Big(d\tau^{d-1} g(\tau x) + \tau^d x \cdot \nabla g(\tau x)\Big)d\tau$$

and, differentiating under the integral in x,

$$x \cdot \nabla(Jg)(x) = d\int_0^1 \tau^{d-1}\big(\tau x \cdot \nabla g(\tau x)\big)d\tau = d\int_0^1 \tau^d\, x \cdot \nabla g(\tau x)\,d\tau.$$

Adding, $d\,(Jg)(x) + x\cdot\nabla(Jg)(x) = d\int_0^1\big(d\tau^{d-1}g(\tau x) + \tau^d x\cdot\nabla g(\tau x)\big)d\tau = d\,g(x)$ by the previous display; dividing by d, $L(Jg)(x) = (Jg)(x) + \frac{x\cdot\nabla(Jg)(x)}{d} = g(x)$, i.e. $LJ = I$. Applying L first and using the same integration with $(Lg)(\tau x) = g(\tau x) + \frac{\tau x\cdot\nabla g(\tau x)}{d}$ gives $JLg = g$; hence $L^{-1} = J$.

(ii) Polar coordinates $x = \rho\theta$, $dx = \rho^{d-1} d\rho\, d\sigma(\theta)$:

$$\|L^{-1}g\|_2^2 = \int_{\mathbb{S}^{d-1}} \int_0^1 \Big| d \int_0^1 \tau^{d-1} g(\tau\rho\theta) d\tau \Big|^2 \rho^{d-1} d\rho\, d\sigma(\theta).$$

Substitute $s = \tau\rho$: $d\int_0^1 \tau^{d-1} g(\tau\rho\theta)d\tau = \frac{d}{\rho^d}\int_0^\rho s^{d-1} g(s\theta)\, ds$. For fixed $\theta \in \mathbb{S}^{d-1}$ set $\gamma(\rho) = g(\rho\theta)$ and equip $(0,1)$ with the measure $d\mu(\rho) = \rho^{d-1}d\rho$, so that $\mu(0,\rho) = \rho^d/d$ and $\|g\|^2_{L^2(D)} = \int_{\mathbb{S}^{d-1}} \|\gamma\|^2_{L^2(d\mu)} d\sigma(\theta)$. The pointwise formula reads

$$(L^{-1}g)(\rho\theta) = \frac{d}{\rho^d}\int_0^\rho \gamma(s)\, d\mu(s) = d\,(H\gamma)(\rho), \qquad (H\gamma)(\rho) := \frac{1}{\mu(0,\rho)}\int_0^\rho \gamma\, d\mu.$$

The averaging operator H is bounded on $L^2(d\mu)$ (Hardy's inequality on $((0,1), d\mu)$). To compute a constant, change variables $t = \rho^d$: then $\rho^{d-1}d\rho = dt/d$, $\mu(0,\rho) = t/d$, and with $\beta(t) := d^{-1/2}\gamma(t^{1/d})$ one has the isometry $\int_0^1 |\beta(t)|^2 dt = \int_0^1 |\gamma(\rho)|^2 \rho^{d-1} d\rho = \|\gamma\|^2_{L^2(d\mu)}$, and

$$(H\gamma)(\rho) = \frac{d}{t}\int_0^{t^{1/d}} \gamma(s)\, s^{d-1} ds = \frac{d}{t}\int_0^t \gamma(s_1^{1/d})\,\frac{ds_1}{d} = \frac{\sqrt{d}}{t}\int_0^t \beta(s_1)\, ds_1,$$

which is $\sqrt{d}$ times the Cesàro operator in t (of norm $\frac{p}{p-1} = 2$ on $L^2((0,1), dt)$). Therefore

$$\|H\gamma\|^2_{L^2(d\mu)} = \int_0^1 |H\gamma|^2 \frac{dt}{d} = \frac{1}{d}\cdot d\int_0^1 \Big|\frac{1}{t}\int_0^t \beta\Big|^2 dt \le \frac{d}{d}\cdot 4\int_0^1 |\beta|^2 dt = 4\,\|\gamma\|^2_{L^2(d\mu)},$$

so $\|H\gamma\|_{L^2(d\mu)} \le 2\|\gamma\|_{L^2(d\mu)}$. Therefore

$$\|L^{-1}g\|^2_{L^2(D)} = d^2\int_{\mathbb{S}^{d-1}} \|H\gamma\|^2_{L^2(d\mu)} d\sigma \le d^2\cdot 4\int_{\mathbb{S}^{d-1}} \|\gamma\|^2_{L^2(d\mu)} d\sigma = 4d^2\,\|g\|^2_{L^2(D)}, \qquad \|L^{-1}g\|_2 \le 2d\,\|g\|_2.$$

Thus $\|L^{-1}\| \le C_d$ with $C_d = 2d$; any d-dependent constant suffices here, as all constants below are allowed to depend on (α, d). We thus take (ii) in the form $\|L^{-1}g\|_2 \le C_d\|g\|_2$.

(iii) From $\|g\|_2 = \|L^{-1}Lg\|_2 \le C_d\|Lg\|_2$, $\|Lg\|_2 \ge c_d\|g\|_2$. On a dense set (C^1) $J = L^{-1}$ with $\|J\| \le C_d$, so J extends to $\mathcal{B}(L^2(D))$; and $\|Lg\|_2 \le \|g\|_2 + \frac{1}{d}\|x\cdot\nabla g\|_2 \le (1+\frac{1}{d})\|g\|_{H^1}$, so $L \in \mathcal{B}(H^1, L^2)$ (it is *not* bounded on L^2). For the last claim, map $g \in L^2$ to the divergence-form distribution $F = \frac{1}{d}\operatorname{div}(xg) \in H^{-1}$ ($\|F\|_{H^{-1}} \le \frac{1}{d}\|g\|_2$) and apply the Dirichlet Laplacian inverse: $-\Delta^{-1} : H^{-1} \to H^1_0$ is bounded, so $T : L^2 \to H^1_0$ is bounded with $\|Tg\|_{H^1} \lesssim \frac{1}{d}\|g\|_2$; by Poincaré $\|Tg\|_2 \lesssim \|g\|_2$. Finally, for $\psi \in L^2$ and $\phi = -\Delta^{-1}\psi \in H^2 \cap H^1_0$ ($\|\phi\|_{H^2} \lesssim \|\psi\|_2$), $|\langle Tg, \psi\rangle| = \frac{1}{d}|\int_D g\, x\cdot\nabla\phi\, dx| \le \frac{1}{d}\|g\|_{H^{-1}}\|x\cdot\nabla\phi\|_{H^1_0} \lesssim \|g\|_{H^{-1}}\|\psi\|_2$, so $T : H^{-1} \to L^2$ is bounded. On $g \in C^1$ the identity $\frac{1}{d}\operatorname{div}(xg) = Lg$ identifies this extension with the T of Lemma 4.1; in particular Tg always satisfies the very-weak form of $\Delta w = Lg$. The refined scale-localized bound $\|Tg\|_2 \lesssim \delta\|g\|_2$ for packets is Lemma 5.3. □

Remark 4.4. Lemma 4.3 is the analytic crux: by Lemma 4.1, $Tg = -\Delta^{-1}Lg$. On a frequency band $|\cdot| \asymp \delta^{-1}$ the Dirichlet Laplacian inverts with gain δ^2 but the *transport source* L is invertible as an order-zero operator (Lemma 4.3: $L^{-1} = J \in \mathcal{B}(L^2)$ and $\|Lg\|_2 \ge c_d\|g\|_2$), so T gains only *one* power of δ. This is made precise in Lemma 5.3.

Lemma 4.5 (Elliptic estimate for the harmonic part)**.** *Let $F \in H^{-1}(D)$ and decompose the solution $w \in H^1_0(D)$ of $-\Delta w = F$, $w|_{\partial D} = 0$, as $w = U - q$ with $U = (-\Delta_{\mathbb{R}^d})^{-1}F \in H^1_{\mathrm{loc}}$ (fundamental-solution representation on the compactly supported F) and $\Delta q = 0$ in D, $q = U$ on ∂D. Then*

$$\|q\|_{L^2(D)} \;\le\; C_E\,\|U\|_{H^1(D)}, \qquad C_E = C_E(d),$$

and consequently, for $F = \frac{1}{d}\operatorname{div}(xg)$ ($g \in L^2$) and $w = Tg$, $\|q\|_2 \le C''_E\|g\|_2$ with $C''_E = C''_E(d)$.

If moreover $\operatorname{supp} g \subset D_{(\tau)} := \{|x| \le 1-\tau\}$ *for some* $\tau \in (0, \frac{1}{2}]$*, then with* $C'''_E = C'''_E(d)$*,*

$$\|U\|_{H^1(\partial D)} \;\le\; C'''_E\, \tau^{-(d-1)} \|g\|_2,$$

i.e. the harmonic trace stays $O(\|g\|_2)$ *(up to a* τ-fixed *factor) and is not amplified as the source recedes from* ∂D*.*

Proof. Set $\phi := q - U$, the solution of $-\Delta\phi = F$ in $H^1_0(D)$ (since $\Delta q = 0$, $-\Delta U = F$, and $\phi = q - U = 0$ on ∂D); H^1_0 energy theory (Lax–Milgram and Poincaré) gives $\|\phi\|_{H^1} \le C_P \|F\|_{H^{-1}}$. Testing against ϕ,

$$\int_D |\nabla\phi|^2 = \langle F, \phi\rangle_{H^{-1}, H^1_0} \le \|F\|_{H^{-1}} \|\phi\|_{H^1},$$

so $\|\phi\|_{H^1} \le C_P \|F\|_{H^{-1}}$, and $\|q\|_{L^2} \le \|U\|_{L^2} + \|\phi\|_{L^2} \le \|U\|_{L^2} + C_P\|F\|_{H^{-1}} \le C_E \|U\|_{H^1}$ after using $\|F\|_{H^{-1}} \lesssim \|U\|_{H^1}$. For the specific $F = \frac{1}{d}\operatorname{div}(xg)$, $\|U\|_{H^1(D)} \lesssim \|F\|_{H^{-1}} \le \frac{1}{d}\|g\|_2$ (interior elliptic regularity on $D \Subset \frac{3}{2}D$ for $-\Delta_{\mathbb{R}^d} U = F$, plus $\|F\|_{H^{-1}} \le \frac{1}{d}\|g\|_2$ from the divergence form), so $\|q\|_2 \le C''_E \|g\|_2$. For the interior claim, use the representation $U = (-\Delta_{\mathbb{R}^d})^{-1}\big(\frac{1}{d}\operatorname{div}(xg)\big) = -\frac{1}{d}(-\Delta_{\mathbb{R}^d})^{-1}\operatorname{div}(xg)$, i.e. $U(x) = -\frac{1}{d}\int \nabla\Phi(x-y)\cdot y\, g(y)\, dy$ with Φ the fundamental solution, $|\nabla\Phi(z)| \lesssim |z|^{1-d}$ (integration by parts is unnecessary: this is the convolution of the L^2 field yg with the locally-integrable kernel $\nabla\Phi$). For $x \in D\setminus D_{(\tau/2)}$ and $y \in D_{(\tau)}$, $|x-y| \ge \tau/2$, so $|\nabla\Phi(x-y)| \lesssim \tau^{1-d}$ and $|U(x)| \lesssim \tau^{1-d}\|g\|_1 \lesssim \tau^{1-d}\|g\|_2$, and the same with one more derivative, $|\nabla U(x)| \lesssim \tau^{-d}\|g\|_2$. Hence for a fixed interior/annular collar $D \setminus D_{(\tau/2)}$, $\|U\|_{H^1(D\setminus D_{(\tau/2)})} \lesssim \tau^{1-d}\|g\|_2$ and by smooth continuation (interior elliptic regularity on $D_{(\tau/2)}$ for the same equation, applied to the patched field) $\|U\|_{H^1(D)} \lesssim \tau^{1-d}\|g\|_2$; the trace theorem $\|u\|_{L^2(\partial D)} \lesssim \|u\|_{H^1(D)}$ then gives $\|U\|_{L^2(\partial D)} \lesssim \tau^{1-d}\|g\|_2$ with ∇U similarly bounded, so $\|U\|_{H^1(\partial D)} \lesssim \tau^{1-d}\|g\|_2$. The stated exponent $\tau^{-(d-1)}$ is not optimal but suffices: any (τ, d)-dependent constant works, since τ is chosen before δ_0. □

4.5 The forward map: very-weak form, apriori bounds, quadratic remainder

Since $f \in C^\alpha$, $\alpha > 1$, f is C^1 and $1 - r \le f \le 1 + r$ uniformly. We use the very-weak (L^2) formulation: $u_f \in L^2(D)$ satisfies, for all $\phi \in H^2(D) \cap H^1_0(D)$,

$$\int_D u_f\, \operatorname{div}(f\nabla\phi)\, dx = \int_D \phi\, dx. \tag{4}$$

Existence/uniqueness of a very-weak solution follows by transposition: since $f \in C^1$ and $1 - r \le f \le 1 + r$, the operator $-\operatorname{div}(f\nabla\cdot) : H^1_0(D) \to H^{-1}(D)$ is an isomorphism (Lax–Milgram), so for $\psi \in L^2(D)$ the adjoint problem $-\operatorname{div}(f\nabla\phi) = \psi$, $\phi \in H^1_0(D)$, has a unique solution $\phi \in H^2(D) \cap H^1_0(D)$; defining $u_f \in L^2(D)$ by $\langle u_f, \psi\rangle := \int_D \phi\, dx$ (well-defined, linear, bounded in ψ since $\|\phi\|_{H^2} \lesssim \|\psi\|_2$) gives (4) upon $\operatorname{div}(f\nabla\phi) = -\psi$, and uniqueness in L^2 follows from the same isomorphy (if $u \in L^2$ satisfies (4) with the right side 0, the density of the adjoint solutions ψ in L^2 forces $u = 0$). All subsequent L^2/weak estimates use only ellipticity and boundedness via (T6); we never invoke $u_f \in H^1_0(D)$ from (4).

Lemma 4.6 (Apriori estimates)**.** *Uniform in* $f \in \mathcal{F}(\alpha, r)$*: (i)* $\|u_f\|_\infty + \|u_f\|_2 \lesssim 1$*; (ii)* $\|u_f - u_0\|_2 \lesssim \|f - \mathbf{1}\|_2 \le r$*; (iii) more generally* $\|u_f - u_0\|_2 \le C_A \|(f-\mathbf{1})\nabla u_0\|_{H^{-1}(D)}$ *with* $C_A = C_A(d, r)$*, and in particular* $\|u_f - u_0\|_2 \lesssim \|f - \mathbf{1}\|_\infty$*; if moreover* $\operatorname{supp}(f - \mathbf{1}) \subset D'$ *with* $\operatorname{dist}(D', \partial D) > \varepsilon_0$*, then* $\|u_f - u_0\|_2 \lesssim_{\varepsilon_0} \|f - \mathbf{1}\|_{H^1(D')}$*.*

Proof. (i) By the comparison principle with barriers $\frac{u_0}{1\mp r} = \frac{|x|^2 - 1}{2d(1\mp r)}$: since $1 - r \le f \le 1 + r$ and $\operatorname{div}(f\nabla u_f) = 1$, the weak maximum principle for the divergence operator $w \mapsto \operatorname{div}(f\nabla w)$ (with the subsolution/supersolution comparison valid in the weak sense) gives, comparing u_f against the two

radial barriers, $\operatorname{div}\big(f\nabla\frac{u_0}{1+r}\big)=\frac{f}{1+r}\Delta u_0=\frac{f}{1+r}\le 1=\operatorname{div}(f\nabla u_f)$ with $\frac{u_0}{1+r}=0=u_f$ on ∂D, so $u_f\ge\frac{u_0}{1+r}$ in D; symmetrically $\operatorname{div}\big(f\nabla\frac{u_0}{1-r}\big)=\frac{f}{1-r}\ge 1$ gives $u_f\le\frac{u_0}{1-r}$. Hence $\frac{|x|^2-1}{2d(1+r)}\le u_f\le\frac{|x|^2-1}{2d(1-r)}\le 0$ and $|u_f|\le\frac{1}{2d(1-r)}\lesssim 1$ on D. (ii) Testing (4) for u_f and u_0 against $\phi=-\Delta^{-1}\psi$ ($\psi\in L^2$ arbitrary, $\|\phi\|_{H^2}\lesssim\|\psi\|_2$) and subtracting:

$$\langle u_f-u_0,\psi\rangle=-\int_D(f-\mathbf{1})\nabla u_0\cdot\nabla\phi\,dx,$$

so $|\langle u_f-u_0,\psi\rangle|\le\|f-\mathbf{1}\|_2\|\nabla u_0\|_\infty\|\nabla\phi\|_2\lesssim\|f-\mathbf{1}\|_2\|\psi\|_2$, and (ii) follows by duality. (iii) Sharpening the same display, $|\langle u_f-u_0,\psi\rangle|\le\|(f-\mathbf{1})\nabla u_0\|_{H^{-1}}\|\nabla\phi\|_{H^1_0}\le C\|(f-\mathbf{1})\nabla u_0\|_{H^{-1}}\|\psi\|_2$, giving $\|u_f-u_0\|_2\lesssim\|(f-\mathbf{1})\nabla u_0\|_{H^{-1}}$. Now $(f-\mathbf{1})\nabla u_0=(f-\mathbf{1})x/d$. On the full ball $\|(f-\mathbf{1})x/d\|_{H^{-1}}\le\|(f-\mathbf{1})x/d\|_2\lesssim\|f-\mathbf{1}\|_\infty$ (since H^{-1} is weaker than L^2 via the embedding of the source as a divergence). If $\operatorname{supp}(f-\mathbf{1})\subset D'\Subset D$, fix a cut-off $\chi_0\in C_c^\infty(D)$, $\chi_0=1$ on D', and compute $\operatorname{div}\big(\chi_0(f-\mathbf{1})x/d\big)=(f-\mathbf{1})\frac{x\cdot\nabla\chi_0}{d}+\chi_0\big(f-\mathbf{1}+\frac{x\cdot\nabla f}{d}\big)$; this is an H^{-1} function of norm $\lesssim\|f-\mathbf{1}\|_{L^2(D')}+\|\nabla f\|_{L^2(D')}$, and $\|(f-\mathbf{1})x/d\|_{H^{-1}}=\|\chi_0(f-\mathbf{1})x/d\|_{H^{-1}}\lesssim\|f-\mathbf{1}\|_{H^1(D')}$ by Lax–Milgram duality (the map $g\mapsto\operatorname{div}(\chi_0 gx/d)$ is bounded $H^1_0(D)\to H^{-1}(D)$ and for g supported in D', $\chi_0 g=g$). (The gradient term ∇f is present because the conductivity does *not* vanish on ∂D; without the interior-support hypothesis the map $f\mapsto u_f$ is $C^\alpha\mapsto L^2$ bounded only with the additive $C(f-\mathbf{1})$ constant of (ii), which is consistent with – and does not improve – the radius minimax rate.) □

Lemma 4.7 (Quadratic remainder)**.** *For $f=\mathbf{1}+h\in\mathcal{F}(\alpha,r)$, write $u_f-u_0=Th+R(h)$. Then $\|R(h)\|_2\le C\|h\|_\infty\|h\|_2$.*

Proof. From the equations for u_f and u_0, $w:=u_f-u_0$ satisfies $\operatorname{div}(f\nabla w)=-\operatorname{div}(h\nabla u_0)$, $w=0$ on ∂D. The energy identity (test against w, justified since $f\in C^\alpha$, $\alpha>1$ gives $w\in H^1$) gives $(1-r)\|\nabla w\|_2^2\le\int_D f|\nabla w|^2=-\int_D h\nabla u_0\cdot\nabla w\le\|h\|_2\|\nabla u_0\|_\infty\|\nabla w\|_2$, hence $\|\nabla w\|_2\lesssim\|h\|_2$. Since $R(h)=-\Delta^{-1}\operatorname{div}(h\nabla w)$, for any $\psi\in L^2$ with $\phi=-\Delta^{-1}\psi$, $\langle R(h),\psi\rangle=-\int_D h\nabla w\cdot\nabla\phi$, so $|\langle R(h),\psi\rangle|\le\|h\|_\infty\|\nabla w\|_2\|\nabla\phi\|_2\lesssim\|h\|_\infty\|h\|_2\|\psi\|_2$. Duality gives the bound. □

Remark 4.8 ($\alpha>1$ is essential). At the minimax scale $\delta_*\asymp n^{-1/(2\alpha+d+2)}$ the wavelet perturbations used below have $\|h\|_\infty\lesssim\delta_*^\alpha$, so $\|R(h)\|_2/(\delta_*\|h\|_2)\lesssim\delta_*^{\alpha-1}\to 0$ *iff* $\alpha>1$. Thus the linearization is faithful precisely in the regime $\alpha>1$, which is why the hypothesis appears.

5 Degree-one smoothing of the linearization

We now make Remark 4.4 precise with an exact Fourier-side computation, which shows T is an operator of order exactly -1.

Lemma 5.1 (Exact smoothing; Fourier form)**.** *Let $\chi\in C_c^\infty(\mathbb{R}^d)$, $0\le\chi\le 1$, $\chi=1$ on $\{|x|\le\frac12\}$, $\operatorname{supp}\chi\subset\{|x|\le\frac34\}$. For $h\in C^\alpha(D)$ set $\tilde h=h\chi$, extended by 0 to $\mathbb{R}^d$. Then, as tempered distributions,*

$$\widehat{L\tilde h}(\xi)=-\frac{1}{d}\,\xi\cdot\nabla_\xi\widehat{\tilde h}(\xi).\tag{5}$$

Consequently, for a scale-δ wavelet packet *$\tilde h$ and dyadic shells $S_k=\{\xi:|\xi|\in[2^k,2^{k+1})\}$,*

$$\||\xi|^{-2}\widehat{L\tilde h}(\xi)\|_{L^2(S_k)}=\frac{1}{d}\,\||\xi|^{-2}\,\xi\cdot\nabla_\xi\widehat{\tilde h}(\xi)\|_{L^2(S_k)},\qquad \|U\tilde h\|_{L^2(S_k)}\asymp\delta\|\tilde h\|_{L^2(S_k)}\ \text{ on the loaded shell } 2^k\asymp\delta^{-1},\tag{6}$$

the two-sided bound being valid for packets *(the situation in which the lower bound is used below; it is* not *a valid two-sided identity for an arbitrary C^α function, whose $\widehat{\tilde h}$ can be radially homogeneous and radially annihilate $\xi\cdot\nabla_\xi$).*

Proof. Using $\widehat{x_j F}(\xi) = i\,\partial_{\xi_j}\widehat{F}(\xi)$ and $\widehat{\partial_j h}(\xi) = i\xi_j \widehat{h}(\xi)$,

$$\widehat{x_j \partial_j \tilde{h}}(\xi) = i\,\partial_{\xi_j}\big(i\xi_j \widehat{\tilde{h}}(\xi)\big) = -\partial_{\xi_j}\big(\xi_j \widehat{\tilde{h}}(\xi)\big) = -\widehat{\tilde{h}}(\xi) - \xi_j \partial_{\xi_j}\widehat{\tilde{h}}(\xi).$$

Summing over $j = 1, \dots, d$: $\widehat{A\tilde{h}}(\xi) = \frac{1}{d}\sum_j \widehat{x_j \partial_j \tilde{h}}(\xi) = -\frac{1}{d}\big(d\,\widehat{\tilde{h}}(\xi) + \xi \cdot \nabla_\xi \widehat{\tilde{h}}(\xi)\big)$. Hence

$$\widehat{L\tilde{h}}(\xi) = \widehat{\tilde{h}}(\xi) + \widehat{A\tilde{h}}(\xi) = -\frac{\xi \cdot \nabla_\xi \widehat{\tilde{h}}(\xi)}{d}.$$

This proves (5). For the packet corollary (6) of a scale-δ packet $\tilde{h}(x) = \sum_m \varepsilon_m a_m \psi((x - x_m)/\delta)$, one expands $\widehat{\tilde{h}}(\xi) = \delta^d \widehat{\psi}(\delta\xi)\sum_m \varepsilon_m a_m e^{-ix_m\cdot\xi}$ and reads off the loaded-shell gain: $\xi \cdot \nabla_\xi$ differentiates the phase $e^{-ix_m\cdot\xi}$ to bring down a factor $x_m \cdot \xi \asymp |x_m|\delta^{-1}$, and on $|x_m| \geq c_0$ this is $\asymp \delta^{-1}$, so $\|\xi\cdot\nabla_\xi \widehat{\tilde{h}}\|_{L^2(S_k)} \asymp \delta^{-1}\|\widehat{\tilde{h}}\|_{L^2(S_k)}$; dividing by $|\xi|^2 \asymp \delta^{-2}$ gives (6) (with the phase interference controlled by separation and Lemma 5.2). The *lower* side of this packet estimate in the uniform sense relevant for (8) is reproved, with constants, inside Lemma 5.3's proof (the display labelled "Lower bound" there); the present form (6) records the identity content. □

Lemma 5.2 (Almost-orthogonality of separated exponentials)**.** *Let $\Lambda = \{x_m\} \subset \{|x| \leq \frac{1}{2}\}$ be 4δ-separated, $|x_m - x_{m'}| \geq 4\delta$. For $b = (b_m) \in \ell^2(\Lambda)$ and $\mathbf{a}_b(\xi) := \sum_m b_m e^{-ix_m\cdot\xi}$, define the frequency-ball average $\mathcal{M}_\rho(b) := \frac{1}{|B_\rho|}\int_{B_\rho}|\mathbf{a}_b(\xi)|^2 d\xi$ on $B_\rho = \{|\xi| \leq \rho\}$, $\rho = \Theta\delta^{-1}$, Θ a fixed constant. Then, uniformly in δ and the configuration Λ,*

$$\mathcal{M}_\rho(b) \;\lesssim\; \sum_m |b_m|^2, \qquad \text{and} \qquad \mathcal{M}_\rho(\{x_m \cdot (\cdot)\}\, b) \;\lesssim\; \delta^{-2}\sum_m |b_m|^2 \tag{7}$$

where the second statement applies to the weighted coefficients $\widetilde{b}_m = x_m \cdot v\, b_m$ with a fixed unit vector v (so $|\widetilde{b}_m| \leq |b_m||x_m| \leq \frac{1}{2}|b_m|$), and the corresponding phase function is $(x_m \cdot v)e^{-ix_m\cdot\xi}$.

Proof. Write, with $\chi_\rho = \mathbf{1}_{B_\rho}$ and $\check{\chi}_\rho$ its inverse Fourier transform, $\int_{B_\rho}|\mathbf{a}_b|^2 d\xi = \sum_{m,m'} b_m \overline{b_{m'}}\,\check{\chi}_\rho(x_{m'} - x_m)$. The diagonal terms ($m = m'$) give $\sum_m |b_m|^2 \check{\chi}_\rho(0) = |B_\rho|\sum_m |b_m|^2$ since $\check{\chi}_\rho(0) = |B_\rho|$. For the off-diagonal part we cannot use the ball indicator's transform (a jinc/Bessel profile decaying only as $|y|^{-(d+1)/2}$), whose Schur sum diverges; instead dominate by a *smooth majorant*: choose $\varphi_0 \in C_c^\infty$ with $\varphi_0(\eta) \geq \mathbf{1}_{|\eta|\leq 1}$ and $|\check{\varphi_0}(y)| \lesssim (1 + |y|)^{-N}$ for any N (a Schwartz function since φ_0 is smooth compactly supported), set $\varphi_\rho(\xi) = \varphi_0(\xi/\rho)$, and note $\check{\varphi_\rho}(y) = \rho^d \check{\varphi_0}(\rho y)$ with $\check{\varphi_\rho}(0) = \rho^d\check{\varphi_0}(0) \asymp |B_\rho|$ and $|\check{\varphi_\rho}(y)| \lesssim \rho^d(1+\rho|y|)^{-N}$. Because $\mathbf{1}_{B_\rho} \leq \varphi_\rho$, $\int_{B_\rho}|\mathbf{a}_b|^2 d\xi \leq \int \varphi_\rho |\mathbf{a}_b|^2 d\xi = \sum_{m,m'} b_m \overline{b_{m'}}\check{\varphi}_\rho(x_{m'} - x_m)$, and the off-diagonal part is bounded by

$$\sum_{m \neq m'} |b_m||b_{m'}|\,\rho^d\big(1 + \rho|x_{m'} - x_m|\big)^{-N} \lesssim \rho^d \sum_{m,m'} \frac{|b_m||b_{m'}|}{(1 + |x_{m'} - x_m|/\delta)^N},$$

using $\rho = \Theta\delta^{-1}$ and $|x_{m'} - x_m| \geq 4\delta$. The kernel $K(x_{m'}, x_m) = (1+|x_{m'} - x_m|/\delta)^{-N}$ on the separated set is Schur-/Cotlar-summable: $\sup_{m'}\sum_m K \lesssim \sum_{k\geq 1} k^{d-1}(1+4k)^{-N} < \infty$ for $N > d$, and symmetrically in m'; hence the off-diagonal quadratic form is $\lesssim \rho^d \sum_m |b_m|^2$. Dividing by $|B_\rho| \asymp \rho^d$ gives the first bound of (7). The second (weighted) bound is the first applied to the coefficients $b_m^{(j)} = a_m x_{m,j}$ for each coordinate j: pointwise, $|\sum_m a_m(x_m \cdot \xi)e^{ix_m\cdot\xi}|^2 = |\sum_j \xi_j \sum_m a_m x_{m,j} e^{ix_m\cdot\xi}|^2 \leq |\xi|^2 \sum_j |\sum_m a_m x_{m,j} e^{ix_m\cdot\xi}|^2$, and averaging each coordinate term over B_ρ with (i), using $|x_{m,j}| \leq |x_m| \leq \frac{1}{2}$ and $|\xi| \leq \rho$,

$$\frac{1}{|B_\rho|}\int_{B_\rho}\Big|\sum_m a_m(x_m \cdot \xi)e^{ix_m\cdot\xi}\Big|^2 d\xi \leq \rho^2 \sum_j \mathcal{M}_\rho(a\, x_{\cdot,j}) \lesssim \rho^2 \sum_j \sum_m a_m^2 x_{m,j}^2 \leq \frac{\rho^2}{4}\sum_m a_m^2 \asymp \delta^{-2}\sum_m a_m^2.$$

This proves the second bound of (7). □

Lemma 5.3 (Scale estimate for wavelet packets)**.** *Fix χ as above and $R \in (0, \frac{1}{8}]$. There are $c_1, C_1 > 0$ (depending only on d, χ, R) such that the following holds. Let $g = \sum_m \varepsilon_m a_m \psi\big(\frac{\cdot - x_m}{\delta}\big)$ be a wavelet packet built from a fixed $\psi \in C_c^\infty$ with $\operatorname{supp}\psi \subset B(0,R)$, $\psi \neq 0$, $\int \psi\, dx = 0$, where $\varepsilon_m \in \{-1, 0, 1\}$, $a_m \in \mathbb{R}$, and the centres x_m lie in the* annulus *$c_0 \le |x_m| \le \frac{1}{2} - c_0$ for a fixed constant $c_0 = c_0(d) \in (0, \frac{1}{16}]$ with pairwise separation $|x_m - x_{m'}| \ge 4\delta$ (so the terms have pairwise disjoint supports, all inside $\{|x| \le \frac{1}{2} - c_0/2\}$, at distance $\ge c_0/2 + R\,\delta \ge c_0/2$ from ∂D; the support stays in $\{|x| \le \frac{3}{4}\}$ too). Then for $\delta \in (0, \delta_0]$,*

$$c_1\,\delta\,\|g\|_{L^2(D)} \;\le\; \|Tg\|_{L^2(D)} \;\le\; C_1\,\delta\,\|g\|_{L^2(D)}, \tag{8}$$

with $\|g\|_2$ computed on D (equivalently $\mathbb{R}^d$, by the support).

Proof. Since $\operatorname{supp} g \subset \{|x| \le \frac{3}{4}\}$ and $\chi = 1$ there, $g = g\chi$, so $\tilde g = g$. Write $Tg = U - q$ with $U = -\Delta_{\mathbb{R}^d}^{-1}(Lg)$ and q harmonic in D, $q = U$ on ∂D.

Whole-space term U. We use the exact identity (5): with $\Lambda F = \xi \cdot \nabla_\xi F$ the dilation generator, $U = \frac{1}{d}|D|^{-2}\Lambda L_{\mathrm{op}}\, g$ is a nonlocal Fourier operator of order exactly -1, and on a scale-δ packet it multiplies the L^2 norm by $\asymp \delta$ (the content of (6)). For concreteness and constants we compute directly. Write $\widehat g(\xi) = \delta^d \widehat\psi(\delta\xi)\mathbf{a}(\xi)$ with $\mathbf{a}(\xi) := \sum_m \varepsilon_m a_m e^{-ix_m\cdot\xi}$. Since the packet terms are disjointly supported, $\|g\|_2^2 = \sum_m a_m^2 \delta^d \|\psi\|_2^2$. With dilation generator $\Lambda F = \xi\cdot\nabla_\xi F$ applied to $F = \widehat g$, and splitting, $\xi\cdot\nabla_\xi \widehat g = \delta^d\big(B_1(\xi) + B_2(\xi)\big)$ with $B_1(\xi) = (\Lambda\widehat\psi)(\delta\xi)\,\mathbf{a}(\xi)$, $B_2(\xi) = \widehat\psi(\delta\xi)\,\xi\cdot\nabla_\xi\mathbf{a}(\xi)$, and $\xi\cdot\nabla_\xi\mathbf{a}(\xi) = -i\sum_m \varepsilon_m a_m (x_m\cdot\xi) e^{-ix_m\cdot\xi}$. Cover the loaded shell $S_k = \{|\xi| \asymp \delta^{-1}\}$ by $O(1)$ balls B_ρ, $\rho \asymp \delta^{-1}$, and apply Lemma 5.2 (claims (i) and the weighted corollary): $\frac{1}{|B_\rho|}\int_{B_\rho}|\mathbf{a}|^2 \lesssim \sum a_m^2$ and $\frac{1}{|B_\rho|}\int_{B_\rho}|\xi\cdot\nabla_\xi\mathbf{a}|^2 \lesssim \rho^2\cdot\frac14\sum a_m^2 \asymp \delta^{-2}\sum a_m^2$. Since $|\widehat\psi(\delta\xi)|, |(\Lambda\widehat\psi)(\delta\xi)| \lesssim$ const, squaring and integrating gives, per contributing term,

$$\begin{aligned}
\int_{S_k} |\xi|^{-4}|\xi\cdot\nabla_\xi\widehat g|^2 d\xi &\lesssim \delta^{2d+4}\Big[\int_{S_k}|B_1|^2 + \int_{S_k}|B_2|^2\Big] \\
&\lesssim \delta^{2d+4}\cdot|S_k|\,\frac{1}{|B_\rho|}\Big[\int_{B_\rho}|\mathbf{a}|^2 + \int_{B_\rho}|\xi\cdot\nabla_\xi\mathbf{a}|^2\Big] \\
&\lesssim \delta^{2d+4}\cdot\delta^{-d}\Big(\sum a_m^2 + \delta^{-2}\sum a_m^2\Big) \;\lesssim\; \delta^2\cdot\delta^d\sum a_m^2 \;=\; C\,\delta^2\,\frac{\|g\|_2^2}{\|\psi\|_2^2}.
\end{aligned} \tag{9}$$

Here the power count is: δ^{2d} (prefactor) $\times\delta^4$ ($|\xi|^{-4}$) $\times\delta^{-d}$ (shell volume) $\times\delta^{-2}$ (weighted AO, dominant B_2 term) $= \delta^2$; the B_1 term is one power of δ smaller. The constant is uniform in M since there are $O(1)$ balls.

For the off-shell shells S_j: those with $2^j \gg \delta^{-1}$ use the Schwartz decay $|\widehat\psi(\delta\xi)| \lesssim (\delta|\xi|)^{-N}$ so each contributes $\delta^2\cdot 2^{-N(j-k)}$; those with $2^j \ll \delta^{-1}$ use the mean-zero bound $|\widehat\psi(\delta\xi)| \lesssim \delta|\xi|$ ($\int\psi = 0$), contributing $\delta^2\cdot 2^{-2(k-j)}$. Both tails sum geometrically, so the whole-space term satisfies the upper bound

$$\|U\|^2_{L^2(\mathbb{R}^d)} \;\lesssim\; \delta^2\|g\|_2^2, \qquad \text{uniformly in } M, \delta. \tag{10}$$

Lower bound (loaded-shell diagonal dominance). Retain only the loaded shell $S_k = \{|\xi| \asymp \delta^{-1}\}$ (all terms non-negative on the Fourier side). Since $\|U\|_2^2 \ge \int_{S_k}|\xi|^{-4}|\widehat{Lg}|^2 d\xi \asymp \delta^4\int_{S_k}|\widehat{Lg}|^2 d\xi$ and by (5) $\widehat{Lg} = -\frac{1}{d}\xi\cdot\nabla_\xi\widehat g$, using $|a+b|^2 \ge \frac12|b|^2 - |a|^2$ with $b = \delta^d\widehat\psi(\delta\xi)\xi\cdot\nabla_\xi\mathbf{a}(\xi)$ and $a = \delta^d(\Lambda\widehat\psi)(\delta\xi)\mathbf{a}(\xi)$:

$$\|U\|_2^2 \ge \frac{\delta^4}{2d^2}\int_{S_k}\delta^{2d}|\widehat\psi(\delta\xi)|^2|\xi\cdot\nabla_\xi\mathbf{a}(\xi)|^2 d\xi - \frac{\delta^4}{d^2}\int_{S_k}\delta^{2d}|(\Lambda\widehat\psi)(\delta\xi)|^2|\mathbf{a}(\xi)|^2 d\xi.$$

Choose once and for all the base profile ψ of the lemma statement: a radial $\psi \in C_c^\infty$ with $\operatorname{supp}\psi \subset B(0,R)$, $\int\psi = \widehat\psi(0) = 0$, and $\widehat\psi$ nonvanishing on the closed annulus $\{1 \le |\eta| \le 2\}$ (readily constructed by applying a power of $-\Delta$ to a radial C_c^∞-bump and adding a small correction). Thus $|\widehat\psi(\delta\xi)| \ge c_\psi > 0$

on the loaded shell while $|(\Lambda\widehat{\psi})(\delta\xi)| \le C_\psi$. Now $\xi\cdot\nabla_\xi \mathbf{a}(\xi) = -i\sum_m \varepsilon_m a_m(x_m\cdot\xi)e^{-ix_m\cdot\delta\xi}$; because the centres lie in the annulus $|x_m| \ge c_0$, on the shell $|x_m\cdot\xi| \asymp |\xi||x_m| \ge \frac{c_0}{2}\delta^{-1}$ (in the diagonal/phase-gradient part). Apply Lemma 5.2's weighted corollary with the *lower* direction: the oscillatory off-diagonals are Schur-summable (by the smooth majorant bound of the lemma) and are dominated by the diagonal, so

$$\int_{S_k} |\widehat{\psi}(\delta\xi)|^2 |\xi\cdot\nabla_\xi \mathbf{a}(\xi)|^2 d\xi \;\ge\; c_\psi^2\,\delta^{-d}\,\delta^{-2}\,\frac{c_0^2}{4}\sum_m a_m^2 \quad \big(\text{diagonal } m=m'\big),$$

the $m=m'$ terms giving $\sum_m a_m^2 \int_{S_k}|x_m\cdot\xi|^2 d\xi \asymp \sum_m a_m^2\delta^{-d}\delta^{-2}$ since $\int_{S_k}|x_m\cdot\xi|^2 d\xi \asymp |x_m|^2\delta^{-2-d}\cdot\delta^2 = |x_m|^2\delta^{-d} \gtrsim c_0^2\delta^{-d}$. Similarly the second (subtracted) term is bounded above by the *unweighted* AO: $\le C_\psi\delta^{-d}\sum a_m^2$ (no δ^{-2}). Hence

$$\|Ug\|_2^2 \;\ge\; \delta^{4+2d}\,\delta^{-d}\Big(\frac{c_\psi^2 c_0^2}{8d^2}\,\delta^{-2}\sum a_m^2 - \frac{C_\psi}{d^2}\sum a_m^2\Big) \;\ge\; c\,\delta^{d+4}\cdot\delta^{-2}\sum a_m^2 \;=\; c\,\delta^2\cdot\delta^d\sum a_m^2 \;=\; c\,\delta^2\,\frac{\|g\|_2^2}{\|\psi\|_2^2},$$

using $\delta^{4+2d}\delta^{-d}\delta^{-2} = \delta^{d+2} = \delta^2\cdot\delta^d$, and for $\delta \le \delta_0(c_\psi, c_0, C_\psi)$ the dominant δ^{-2} term controls the subtracted one. Hence $\|U\|_2 \ge c'_U\,\delta\|g\|_2$ with $c'_U = c'_U(d,\psi,c_0) > 0$.

Harmonic correction q. By Lemma 4.5 (applied with $F = Lg = \frac{1}{d}\operatorname{div}(xg)$, for which the solution $w = Tg$ splits as $w = U - q$ with exactly the present U and q), $U = (-\Delta_{\mathbb{R}^d})^{-1}F$ is a convolution with a locally-integrable kernel of a field supported in $D_{(\tau_0)}$, $\tau_0 = \frac{1}{8}$, and on ∂D one has the interior bound $\|U\|_{L^\infty(\partial D)} \lesssim \|g\|_2$ (Cauchy–Schwarz in y, distance $\ge \tau_0/4$ to the support). Split $q = q_0 + q_{\mathrm{osc}}$ with $q_0 = |\partial D|^{-1}\int_{\partial D} q$ the constant Poisson mode; the maximum principle gives $\|q_0\|_{L^2} \le |\partial D|^{-1}\int_{\partial D}|U| \lesssim \delta\|g\|_2$ by the bump-wise multipole expansion of $U|_{\partial D}$ (monopole killed by $\int\psi = 0$, first-order remainder $O(\delta\|\psi_m\|_1)$, $\sum_m |a_m|\|\psi_m\|_1 \lesssim \|g\|_2$), and the mean-zero part decays into the interior at the Poisson rate: $q_{\mathrm{osc}} = \sum_\Lambda \rho^{\sqrt{\Lambda}}\widehat{U}(\Lambda)$, so on $\{|x| \le \frac{3}{4}\}$ (which carries the shell mass of U) $\|q_{\mathrm{osc}}\|_{L^2} \le \big(\max_\Lambda \rho^{2\sqrt{\Lambda}}\big)^{1/2}\|U\|_{L^2(\partial D)} \lesssim e^{-c/\delta}\|g\|_2 \ll \delta\|g\|_2$ (the largest $\sqrt{\Lambda}$ is $\asymp \delta^{-1}$, and $\rho^{2\sqrt{\Lambda}} \le (\frac{3}{4})^{2c\delta^{-1}} \le e^{-c'/\delta}$). Hence

$$\|q\|_{L^2(|x|\le\frac{3}{4})} \;\lesssim\; \delta\,\|g\|_2, \qquad \text{constant depending on } (d,\psi,c_0,\tau_0), \tag{11}$$

$\{|x| \le \frac{3}{4}\}$ being where the shell mass of U lies ($\operatorname{supp} g \subset \{|x| \le \frac{3}{4}\}$, the $\mathbb{R}^d$ integral dominated by its $\{|x| \le \frac{3}{4}\}$ part up to the $e^{-c/\delta}$ tail already accounted).

Combining, $\|Tg\|_2 \le \|U\|_2 + \|q\|_2 \lesssim \delta\|g\|_2$ and $\|Tg\|_2 \ge \|U\|_{L^2(|x|\le\frac{3}{4})} - \|q\|_{L^2(|x|\le\frac{3}{4})} \ge (c'_U/2 - C_q\delta)\,\delta\|g\|_2$ for the $\{|x| \le \frac{3}{4}\}$-restricted shell lower bound $c'_U/2$ (same argument, all estimates on the shrunk domain) and C_q the constant of (11); with δ_0 shrunk so $C_q\delta_0 \le c'_U/4$,

$$\|Tg\|_2 \;\ge\; \tfrac{c'_U}{4}\,\delta\|g\|_2 =: c_1\delta\|g\|_2, \qquad \|Tg\|_2 \;\le\; (C_U + C_q)\delta\|g\|_2 =: C_1\delta\|g\|_2,$$

with c_1, C_1, δ_0 depending only on (d,ψ,c_0). (The origin-centre modes, excluded here by the annulus hypothesis, are treated in Lemma 7.2 via the exact radial computation.) □

Remark 5.4 (Constant bookkeeping in (8)). Both $C''_E = C''_E(d)$ (trace/elliptic constant of the unit ball) and $c'_U = c'_U(d,\psi,c_0)$ are (α,n,δ,r)-independent, so the threshold $\delta_0 = \delta_0(c_\psi, c_0, C_\psi, C''_E)$ and all constants below keep the (α,d,ψ)-dependence only; boundary-near bumps are *not* asked to be small, the loss being absorbed by shrinking δ_0.

Remark 5.5 (T is a degree-one operator, globally). For a full wavelet expansion $h = \sum_{j\ge j_0} h_j$ with h_j at scale 2^{-j}, applying Lemma 5.3 to each dyadic piece and summing (the pieces are almost orthogonal in L^2, and T maps the scale 2^{-j} piece into functions still almost-orthogonal up to the operator's pseudo-local tail) yields $\|Th\|_2^2 \lesssim \sum_j 2^{-2j}\|h_j\|_2^2 \lesssim \|h\|_{H^{-1}}^2$, i.e. T maps H^s into H^{s+1} type spaces with exactly one derivative of smoothing, globally on the Besov scale. In particular $\|Th\|_2 \le C\|h\|_2$, the only global fact (besides (8)) used below.

6 Lower bound

Proposition 6.1. *There exist $r_0 = r_0(\alpha, d) \in (0, 1]$ and $c > 0$ such that for all $r \in (0, r_0]$ and all large n,*

$$\inf_{\widehat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha, r)} \mathbb{E}_{f_0} \|\widehat{f}_n - f_0\|_2^2 \;\geq\; c\, n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

Proof. **Step 1 (wavelet packet).** Fix the single base profile $\psi \in C_c^\infty(\mathbb{R}^d)$ of Lemma 5.3 (radial, $\operatorname{supp}\psi \subset \{|x| \leq \frac{1}{4}\}$, $\int \psi = \widehat{\psi}(0) = 0$, $\widehat{\psi}$ nonvanishing on $\{1 \leq |\eta| \leq 2\}$), normalized so $\|\psi\|_{C^\alpha} = 1$; then $\|\psi\|_2 > 0$ and $\|\psi\|_{C^\alpha} = 1$ are fixed (α, d)-constants. For $\delta \in (0, \delta_0]$ choose a maximal collection $\{x_m\}_{m=1}^M \subset \{c_0 \leq |x| \leq \frac{1}{2}\}$ with pairwise distance $\geq 4\delta$ (hence $M \asymp \delta^{-d}$), and set $\psi_m(x) = \psi\big(\frac{x - x_m}{\delta}\big)$. Then $\operatorname{supp}\psi_m \subset B(x_m, \delta/4)$, $\|\psi_m\|_\infty = \|\psi\|_\infty$, $\|\psi_m\|_2 = \delta^{d/2}\|\psi\|_2$, $\|\psi_m\|_{C^\alpha} = \delta^{-\alpha}\|\psi\|_{C^\alpha}$ (the C^α norm is $\asymp \delta^{-\alpha}$ for $\delta \leq 1$: derivatives scale as δ^{-j} and the top t-Hölder/Zygmund seminorm as $\delta^{-\alpha}$, dominating the lower-order terms). The supports of the ψ_m are pairwise disjoint and contained in $\{|x| \leq \frac{3}{4}\}$, so every partial sum $\sum_{m \in S} \pm\psi_m$ is a wavelet packet of the form treated in Lemma 5.3 (fixed base profile ψ, centres in $\{c_0 \leq |x| \leq \frac{1}{2}\}$, separation $\geq 4\delta$; the annulus restriction keeps the two-sided scale estimate of Lemma 5.3).

Step 2 (hypotheses). By Varshamov–Gilbert, there is a family $\mathcal{S}$ of subsets $S \subset \{1, \dots, M\}$ with $|S| = M/2$, each pair $S \neq S'$ at Hamming distance $|S \triangle S'| \geq M/8$, and $\log|\mathcal{S}| \geq c_0 M$ for an absolute $c_0 > 0$. Index hypotheses by $S \in \mathcal{S}$ and a sign/scale parameter $\kappa > 0$ (fixed below):

$$f_S = \mathbf{1} + \kappa\delta^\alpha \Psi_S, \qquad \Psi_S := \sum_{m \in S} \psi_m.$$

Because the ψ_m have disjoint supports with the same profile, $\|\Psi_S\|_{C^\alpha} \lesssim \max_m \|\psi_m\|_{C^\alpha} \asymp \delta^{-\alpha}\|\psi\|_{C^\alpha}$: the C^a norm of the sum is the max over the summands (disjoint supports), and the top t-Hölder/Zygmund difference quotient satisfies $\sup_{x,y} |\nabla^a \Psi_S(x) - \nabla^a \Psi_S(y)|/|x-y|^t \leq 2^{1-t}\delta^{-\alpha}\|\nabla^a\psi\|_\infty + \delta^{-\alpha}[\psi]_{\dot{C}^t}$ — cross-support quotients are the same order as single-bump ones, giving comparability with a ψ-dependent constant $\leq 2^{1-t} + 1$. Hence

$$\|f_S - \mathbf{1}\|_{C^\alpha} \lesssim \kappa\delta^\alpha \cdot \delta^{-\alpha}\|\psi\|_{C^\alpha} = \kappa\|\psi\|_{C^\alpha} \;\leq\; r$$

provided $\kappa \leq r\|\psi\|_{C^\alpha}^{-1}$, i.e. $\kappa \leq r$ since $\|\psi\|_{C^\alpha} = 1$ (Step 1). We take the maximal admissible $\kappa = r$; the separation below is then $\varepsilon^2 \asymp r^2\delta^{2\alpha}$, so the lower-bound constant inherits the factor r^2 of the shrinking class $\mathcal{F}(\alpha, r)$ – a fixed positive constant for every fixed $r \in (0, r_0]$. Thus $f_S \in \mathcal{F}(\alpha, r)$ for all $S \in \mathcal{S}$. Also, by disjoint supports,

$$\|\Psi_S\|_2^2 = |S|\,\|\psi_m\|_2^2 \asymp M\,\delta^d \asymp 1, \qquad \|\Psi_S\|_\infty = \|\psi\|_\infty.$$

Step 3 (separation). For $S \neq S'$, $f_S - f_{S'} = \kappa\delta^\alpha \sum_{m \in S \triangle S'} \pm\psi_m$, the supports disjoint, so

$$\|f_S - f_{S'}\|_2^2 = \kappa^2\delta^{2\alpha}|S \triangle S'|\,\|\psi_m\|_2^2 \asymp \kappa^2\delta^{2\alpha} \cdot M \cdot \delta^d \asymp \kappa^2\delta^{2\alpha}.$$

So the family is ε-separated with $\varepsilon^2 \asymp \kappa^2\delta^{2\alpha}$.

Step 4 (KL control). With $g_S := f_S - \mathbf{1} = \kappa\delta^\alpha\Psi_S$: g_S is supported in $\{|x| \leq \frac{3}{4}\}$, is a single-scale packet of the form in Lemma 5.3 (fixed profile ψ with $\int\psi = 0$, centres in $\{|x| \leq \frac{1}{2}\}$ separated by $\geq 4\delta$; each ψ_m has $\|\nabla\psi_m\|_2 \asymp \delta^{-1}\|\psi_m\|_2$), so by Lemma 5.3,

$$\|Tg_S\|_2 \;\asymp\; \delta\|g_S\|_2 \;\asymp\; \delta \cdot \kappa\delta^\alpha\|\Psi_S\|_2 \;\asymp\; \kappa\delta^{\alpha+1}.$$

(Here $\|\Psi_S\|_2 \asymp 1$ by Step 2.) The quadratic remainder, by Lemma 4.7,

$$\|R(g_S)\|_2 \lesssim \|g_S\|_\infty\|g_S\|_2 = \kappa\delta^\alpha\|\psi\|_\infty \cdot \kappa\delta^\alpha\|\Psi_S\|_2 \asymp \kappa^2\delta^{2\alpha}.$$

Since $\alpha > 1$, $\|R(g_S)\|_2 \lesssim \kappa^2\delta^{2\alpha} \ll \kappa\delta^{\alpha+1} \asymp \|Tg_S\|_2$ for δ small (the ratio is $\kappa\delta^{\alpha-1} \to 0$ — the decisive use of $\alpha > 1$). Therefore

$$\|u_{f_S} - u_0\|_2 \le \|Tg_S\|_2 + \|R(g_S)\|_2 \lesssim \kappa\delta^{\alpha+1},$$

and by (1),

$$\mathrm{KL}(P_{f_S} \,\|\, P_{f_0}) = \frac{n}{2}\|u_{f_S} - u_0\|_2^2 \le Cn\kappa^2\delta^{2\alpha+2}. \tag{12}$$

Step 5 (Fano). Take $\delta = c\, n^{-1/(2\alpha+d+2)}$. Then the balance is

$$n\delta^{2\alpha+2} = c^{2\alpha+2} n^{1-\frac{2\alpha+2}{2\alpha+d+2}} = c^{2\alpha+2} n^{\frac{d}{2\alpha+d+2}} = c^{2\alpha+2}\delta^{-d}.$$

Choosing $c > 0$ small makes $Cn\kappa^2\delta^{2\alpha+2} \le \frac{1}{10}\,\delta^{-d} \le \frac{1}{10} \cdot \frac{\log|\mathcal{S}|}{c_0}$, i.e. the average KL is $\le \frac{1}{10}\log|\mathcal{S}|$. With $\varepsilon^2 \asymp \kappa^2\delta^{2\alpha} = \kappa^2 c^{2\alpha} n^{-\frac{2\alpha}{2\alpha+d+2}}$ and $\log|\mathcal{S}| \ge c_0 M \asymp \delta^{-d}$, Fano's inequality (Lemma 6.2) gives

$$\max_{S\in\mathcal{S}} \mathbb{E}_{f_S}\|\widehat{f} - f_S\|_2^2 \;\ge\; \frac{\varepsilon^2}{8}\Big(1 - \frac{1}{10} - \frac{\log 2}{\log|\mathcal{S}|}\Big) \;\ge\; c'\, n^{-\frac{2\alpha}{2\alpha+d+2}}$$

for n large. Since $\{f_S\} \subset \mathcal{F}(\alpha, r)$, the minimax risk over $\mathcal{F}(\alpha, r)$ is at least this, proving Proposition 6.1. □

Lemma 6.2 (Fano, applied form)**.** *Let $|\Theta| = N \ge 2$, $\{P_\theta\}_{\theta\in\Theta}$ probability measures with $\frac{1}{N}\sum_\theta \mathrm{KL}(P_\theta\|P_0) \le \beta\log N$, $\beta < 1$, and parameter values separated: $\|f_\theta - f_{\theta'}\|_2 \ge \varepsilon$ for $\theta \ne \theta'$. Then*

$$\inf_{\widehat{f}} \max_\theta \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \;\ge\; \frac{\varepsilon^2}{4}\Big(1 - \beta - \frac{\log 2}{\log N}\Big)\cdot\frac{1}{2} \;\ge\; \frac{\varepsilon^2}{8}\Big(1 - \beta - \frac{\log 2}{\log N}\Big).$$

Proof. Given any estimator $\widehat{f}$, let $\widehat{\theta} = \arg\min_{\theta\in\Theta}\|\widehat{f} - f_\theta\|_2$. By ε-separation and the triangle inequality, $\|\widehat{f} - f_\theta\|_2 \ge \varepsilon/2$ whenever $\widehat{\theta} \ne \theta$. Markov gives $\mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge \frac{\varepsilon^2}{4}P_\theta(\widehat{\theta} \ne \theta)$. Averaging and using $\max_\theta \ge \frac{1}{N}\sum_\theta$: $\max_\theta \mathbb{E}_\theta\|\widehat{f} - f_\theta\|_2^2 \ge \frac{\varepsilon^2}{4}\cdot\frac{1}{N}\sum_\theta P_\theta(\widehat{\theta} \ne \theta)$. Fano's inequality bounds the average testing error below by $1 - \frac{\beta\log N + \log 2}{\log N} = \big(1 - \beta - \frac{\log 2}{\log N}\big)$. Combining gives the claim (Tsybakov, *Introduction to Nonparametric Estimation*, Theorem 2.5). □

7 Upper bound

Proposition 7.1. *There is an estimator $\widehat{f}_n$ with*

$$\sup_{f_0\in\mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\|\widehat{f}_n - f_0\|_2^2 \;\le\; C\, n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

The construction: (A) discretize the observation and set up a scale-δ sieve least-squares problem; (B) a discrete inf–sup/smoothing inequality inverting T on the sieve; (C) balance bias, variance, and the quadratic remainder.

> **Reviewer 3**
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> There is no mathematically precise explanation of the objects used, which makes it hard to understand the observation scheme and to construct the estimator.

7.1 (A) Observation discretization; sieve least squares

Let $V_\delta \subset L^2(D)$ be the span of the Dirichlet eigenfunctions of $-\Delta$ with eigenvalue $\le C_W\,\delta^{-2}$ (C_W a fixed large constant chosen in Lemma 7.2), i.e. the eigenfunctions oscillating at frequencies $\lesssim \delta^{-1}$; by Weyl's law $\dim V_\delta \asymp \delta^{-d}$ (wavelength-2^{-j} eigenfunctions carry eigenvalues $\asymp 2^{2j}$, so the cutoff $\lesssim \delta^{-2}$ keeps *physical scale* $\gtrsim \delta$). Let P_δ be the orthogonal projection onto V_δ and set

$$Y_\delta := P_\delta Y = P_\delta u_{f_0} + n^{-1/2}\xi_\delta, \qquad \xi_\delta := P_\delta \mathbb{W}, \quad \mathbb{E}\|\xi_\delta\|_2^2 = \dim V_\delta \asymp \delta^{-d}.$$

The sieve must avoid both degeneracies of the transport field $\nabla u_0 = x/d$: the origin (where it vanishes) and the boundary collar (where the Hölder boundary estimate loses one degree). We therefore estimate only interior content.

> **Reviewer 3**
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> A more precise presentation and discussion of the set $\mathcal{F}_\delta$ and the properties derived from it (see Equations 13 and 14) is necessary in order to understand the approach and verify its correctness.

Let $\mathcal{F}_\delta$ be the span of all wavelet packets at scales $\ge \delta$ *whose centres lie in the fixed interior* $D_1 := \{c_1 \le |x| \le \frac{1}{2}\}$ for a small fixed $c_1 = c_1(d) > 0$, intersected with $\mathcal{F}(\alpha, r)$. Fix a patch ζ, $\equiv 1$ on $\operatorname{supp}(\mathcal{F}_\delta - \mathbf{1})$, $\equiv 0$ on $B_{c_1/2} \cup \{|x| > \frac{3}{4}\}$. For a general $f \in \mathcal{F}(\alpha, r)$ write $f = f_{\le\delta} + f_{>\delta}$ (scales $\ge \delta$ and $< \delta$; $\operatorname{supp}(f_{\le\delta} - \mathbf{1}) \subset \{\zeta = 1\}$) and set

$$\Pi_\delta f \;:=\; \mathbf{1} + \big(f_{\le\delta} - \mathbf{1}\big)\,\zeta, \qquad \text{so } \Pi_\delta f = \mathbf{1} \text{ off } \operatorname{supp}\zeta \text{ (a ball about 0 and a collar of } \partial D). \tag{13}$$

Thus $\Pi_\delta f \in \mathcal{F}(\alpha, 2r)$, and the comparison point $m = \Pi_\delta f_0$ is Dirichlet ($u_m = 0$ on ∂D, as is u_{f_0}, by (4)) and interior-regular. By the standard $B^\alpha_{\infty,\infty} = C^\alpha$ wavelet characterization,

$$\|f_{>\delta}\|_\infty \le C\delta^\alpha, \quad \|f_{>\delta}\|_2 \le C\delta^\alpha, \quad \|f_{\le\delta}\|_{C^\alpha} \le C\|f\|_{C^\alpha}, \quad \dim \mathcal{F}_\delta \lesssim \delta^{-d}. \tag{14}$$

The *bias* $e = \Pi_\delta f - f = -f_{>\delta} + (f_{\le\delta} - \mathbf{1})(\zeta - \mathbf{1})$ has $\|e\|_2 \lesssim \delta^\alpha$ by (14); its dominant content is interior (the origin/boundary remainder is handled in the collar split of (23)). The projection P_δ retains the part of u_{f_0} oscillating at scales $\gtrsim \delta$ and makes the noise finite-dimensional, so the estimator and its risk are well-defined on the white-noise model. Define the sieve least-squares estimator

$$\widehat{f} = \arg\min_{g \in \mathcal{F}_\delta} \|P_\delta u_g - Y_\delta\|_2^2.$$

Since $\mathcal{F}_\delta$ is closed and $g \mapsto \|P_\delta u_g - Y_\delta\|_2$ is continuous and coercive ($\|P_\delta u_g\|_2 \to \infty$ with $\|g\|_2$ on the sieve by (15) and the apriori bound, Lemma 4.6), a measurable minimizer $\widehat{f}$ exists (selection theorem on the closed finite-dimensional sieve); only its risk is analyzed below.

7.2 (B) Discrete inf–sup (scale inversion on the sieve)

Lemma 7.2. *There are $c, C > 0$ such that for all small δ the following holds. Let $h = \sum_\lambda h_\lambda$ be a wavelet-packet expansion where each h_λ is a packet of Lemma 5.3-type with centre x_λ and scale $\delta_\lambda \ge \delta$ (on the sieve of §(A), every packet has its centre in the interior annulus, so the subfamily below* is *the whole sieve error). For the subfamily with centres in the annulus $c_0 \le |x_\lambda| \le \frac{1}{2}$ (notation $h^{\mathrm{bulk}} = \sum_{\lambda:\, c_0 \le |x_\lambda| \le \frac{1}{2}} h_\lambda$),*

$$c\,\delta\|h^{\mathrm{bulk}}\|_2 \;\le\; \|P_\delta T h^{\mathrm{bulk}}\|_2 \;\le\; C\,\delta\|h^{\mathrm{bulk}}\|_2, \qquad \textit{and more globally} \qquad \|Th\|_2 \le C\|h\|_2. \tag{15}$$

For origin-near modes ($|x_\lambda| \le \rho\delta_\lambda$ for a fixed ρ) the transport field degenerates and the gain is only $\gtrsim \delta_\lambda^2$ (Lemma 7.3); they are excluded *from the sieve of §(A) and projected into the bias, see Remark 7.4.*

Lemma 7.3 (Exact radial gain near the origin)**.** *Let $g(x) = \varphi(|x|/\delta)$ be radial, supported in B_δ about the origin, $\|g\|_2 \asymp \delta^{d/2}$. Then the solution $w = Tg$ of the radial part of (3) satisfies $w(x) = -\frac{1}{d}\int_{|x|}^{\delta} s\, g(s)\, ds$ for $|x| \le \delta$, $w \equiv 0$ for $|x| \ge \delta$, and $\|Tg\|_2 \asymp \delta^2 \|g\|_2$. More generally, for a mode centred at distance $|x_\lambda|$ with scale δ_λ, the mode-wise gain is at least the radial endpoint: for an origin-near mode ($|x_\lambda| \le \rho\, \delta_\lambda$, ρ fixed),*

$$\|Th_\lambda\|_2 \;\gtrsim\; \delta_\lambda^2\, \|h_\lambda\|_2. \tag{16}$$

(used in Remark 7.4 to record the degeneracy).

Proof of Lemma 7.3. For the radial bump, (3) reads $\Delta w = Lg = \frac{1}{d}\operatorname{div}(xg)$, i.e. $\frac{1}{r^{d-1}}\partial_r(r^{d-1}w') = \frac{1}{d}\frac{1}{r^{d-1}}\partial_r(r^d g)$, so $r^{d-1}w'(r) = \frac{1}{d}r^d g(r) + C$; regularity at $r = 0$ forces $C = 0$, so $w'(r) = \frac{r}{d}g(r)$ for all $r > 0$. On $[\delta, 1]$ the source $g \equiv 0$, hence $w' = 0$ there, and the boundary condition $w(1) = 0$ gives $w \equiv 0$ on $[\delta, 1]$. Continuity of w at $r = \delta$ (the flux w' is bounded) then yields $w(x) = -\frac{1}{d}\int_{|x|}^{\delta} s\, g(s)\, ds$ for $|x| \le \delta$, and

$$\|w\|_2^2 = |\mathbb{S}^{d-1}| \int_0^\delta \Big|\frac{1}{d}\int_r^\delta s g(s)\, ds\Big|^2 r^{d-1} dr \;\asymp\; \delta^4 \|g\|_2^2$$

by Cauchy–Schwarz (both sides $\asymp \delta^4\|g\|_2^2$, e.g. the lower bound from profiles with mass near $s = \delta$ and the upper from $|\int_r^\delta sg| \le \delta\|g\|_{L^2(r,\delta)}$ and the δ-length). Thus $\|Tg\|_2 \asymp \delta^2\|g\|_2$.

For the one-sided bound (16), re-run Lemma 5.3's loaded-shell power count; the *only* change is the shift factor $|x_m|^2$ in the weighted almost-orthogonality. With that proof's notation ($\delta = \delta_\lambda$; shell $S_k = \{|\xi| \asymp \delta_\lambda^{-1}\}$, volume $\asymp \delta_\lambda^{-d}$; $\rho_k \asymp \delta_\lambda^{-1}$), the shell integrand $|\xi|^{-4}|\widehat{Lg}|^2$ has its dominant $\xi \cdot \nabla_\xi \mathbf{a}$ factor bounded by $|S_k|\rho_k^2 \sum_m |x_m|^2 a_m^2$ (Lemma 5.2, weighted part). Annulus centres have $|x_m| \asymp 1$; origin-near centres have $|x_m| \lesssim \delta_\lambda$, so $|x_m|^2 \lesssim \delta_\lambda^2$ contributes one full extra power of δ_λ: the count is

$$\|U\|_2^2 \;\gtrsim\; \underbrace{\delta_\lambda^4}_{|\xi|^{-4}}\ \underbrace{\delta_\lambda^{2d}}_{\text{prefactor}}\ \underbrace{\delta_\lambda^{-d}}_{|S_k|}\ \underbrace{\rho_k^2}_{\asymp\delta_\lambda^{-2}}\ \underbrace{\delta_\lambda^2}_{|x_m|^2} \sum_m a_m^2 \;\asymp\; \delta_\lambda^4\,(\delta_\lambda^d \sum_m a_m^2) \;=\; \delta_\lambda^4 \|h_\lambda\|_2^2,$$

using $\|h_\lambda\|_2^2 \asymp \delta_\lambda^d \sum a_m^2$. The subtracted $(\Lambda\widehat{\psi})$-term lacks the ρ_k^2 amplification and is one power smaller, as in Lemma 5.3; the harmonic correction $q = O(\delta_\lambda\|h_\lambda\|_2)$ by (11) absorbs into δ_0. Hence $\|Th_\lambda\|_2 \gtrsim \delta_\lambda^2\|h_\lambda\|_2$ (constants depending on (d, ψ, c_0, ρ) only). □

Proof of Lemma 7.2. We prove the two-sided estimate for the bulk subfamily (the whole sieve error on the interior sieve of §(A)) and the global bound; the origin modes it excludes are projected into Π_δ (Remark 7.4). Throughout, P_δ is the spectral projection of the Dirichlet Laplacian onto $\lambda_\nu \le C_W\, \delta^{-2}$ (spatial frequencies $\lesssim \delta^{-1}$, matching the one-degree smoothing; see §(A)).

(i) Lower bound on $\|Th^{\mathrm{bulk}}\|_2$. Decompose $h^{\mathrm{bulk}} = \sum_j h_j$ into wavelet pieces at scales $2^{-j} \ge \delta$ with centres in the annulus (each h_j satisfies the hypotheses of Lemma 5.3 with scale $\delta_j = 2^{-j}$ and annulus centres). By Lemma 5.3's lower bound applied per scale, $\|Th_j\|_2 \gtrsim 2^{-j}\|h_j\|_2$, and the cross-terms $\langle Th_j, Th_{j'}\rangle$ for $j \ne j'$ are almost-orthogonal by Lemma 5.2 (each Th_j remains concentrated near scale 2^{-j}, both by the Fourier localization of the source and by the Schur-summable interference off the diagonal); hence

$$\|Th^{\mathrm{bulk}}\|_2^2 \;\gtrsim\; \sum_j 2^{-2j}\|h_j\|_2^2 \;\ge\; \delta^2 \sum_j \|h_j\|_2^2 \;\asymp\; \delta^2 \|h^{\mathrm{bulk}}\|_2^2,$$

the last equivalence by the square-function characterization (T5). Hence $\|Th^{\mathrm{bulk}}\|_2 \ge c\delta\|h^{\mathrm{bulk}}\|_2$.

(ii) Spectral tail $\|(I - P_\delta)Th\|_2$. The projector cuts at $|\xi| \asymp \sqrt{C_W}\, \delta^{-1}$, retaining scales $\gtrsim \delta$; the mass of Th^{bulk} beyond this is the off-shell tail of Lemma 5.3. For a single scale-δ packet, (10)'s off-shell shells

S_j, $2^j \gg \delta^{-1}$, contribute a factor $2^{-N(j-k)}$ each via the Schwartz decay $|\widehat{\psi}(\delta\xi)| \lesssim (\delta|\xi|)^{-N}$ (any fixed N), so on $\{|\xi| \gtrsim \sqrt{C_W}\delta^{-1}\}$ the mass is

$$\|P_{\{|\xi|\gtrsim\sqrt{C_W}\delta^{-1}\}}\,U\|_2^2 \;\lesssim \sum_{j:\,2^j\gtrsim\sqrt{C_W}\delta^{-1}} 2^{-N(j-k)}\,\delta^2\|g\|_2^2 \;\lesssim\; C_W^{-N/2}\,\delta^2\|g\|_2^2,$$

(geometric sum over off-shell shells, $2^{j-k} \gtrsim \sqrt{C_W}$). The harmonic correction q is smooth into any Sobolev norm with constant $O(\delta\|g\|_2)$ by (11), so its high tail is Schwartz-small and absorbs into the same bound. For coarser pieces $2^{-j} \in [\delta, 1]$ the cutoff $\sqrt{C_W}\delta^{-1}$ sits even deeper in the off-shell tail, so the bound is uniform; summing over scales as in step (i),

$$\|(I - P_\delta)Th^{\mathrm{bulk}}\|_2 \;\le\; \widetilde{C}\,C_W^{-N/4}\,\delta\|h^{\mathrm{bulk}}\|_2, \tag{17}$$

with $\widetilde{C} = \widetilde{C}(d, \psi, c_0)$. Choosing the fixed constant C_W (hence the sieve V_δ) so large that $\widetilde{C}\,C_W^{-N/4} \le c/4$ ($N \ge 4$ fixed, C_W independent of δ) makes the tail $\le \frac{c}{4}\delta\|h^{\mathrm{bulk}}\|_2$.

(iii) Combination. By the triangle inequality (ordinary norm subtraction),

$$\|P_\delta Th^{\mathrm{bulk}}\|_2 \;\ge\; \|Th^{\mathrm{bulk}}\|_2 - \|(I - P_\delta)Th^{\mathrm{bulk}}\|_2 \;\ge\; c\,\delta\|h^{\mathrm{bulk}}\|_2 - \tfrac{c}{4}\,\delta\|h^{\mathrm{bulk}}\|_2 \;=\; \tfrac{3c}{4}\,\delta\|h^{\mathrm{bulk}}\|_2,$$

combining step (i) and (17); both terms are $O(\delta\|\cdot\|_2)$ and the tail constant $c/4 < c$, so the difference is positive. The upper bound $\|P_\delta Th^{\mathrm{bulk}}\|_2 \le \|Th^{\mathrm{bulk}}\|_2 \lesssim \delta\|h^{\mathrm{bulk}}\|_2$ and the global $\|Th\|_2 \le C\|h\|_2$ follow from Lemma 5.3 / Remark 5.5. □

Remark 7.4 (Origin content and the interior sieve). Lemma 7.3 records that the transport field's degeneracy makes the origin the *least* revealing region (gain only $\asymp \delta_\lambda^2$ there, versus $\asymp \delta_\lambda$ in the bulk annulus). Since an adversarial h could concentrate at the origin — where inversion against the $O(\delta)$ noise/bias terms would lose a factor δ^{-1} — the sieve $\mathcal{F}_\delta$ is restricted to bulk-annulus packets, and the origin/boundary content is projected into $\Pi_\delta f_0$ (hence into the bias, whose u-image is handled by the degree-one gain of (23)). Accordingly, in §(C) the lower bound of (15) is applied to the *whole* sieve error h, supported in the bulk annulus by construction, and the analysis uses the annulus two-sided estimate of Lemma 5.3, valid for all $\alpha > 1$, $d \ge 2$ (Lemmas 5.3; c_0, c_ψ, C_E'' absorbed).

> **Reviewer 3**
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> Although Lemma 7.2 and 7.3 recognize that the estimation problem becomes more difficult near the origin, Remark 7.4 does not adequately address this issue. The results are used in Equation 19, which then leads to incorrect convergence rates.

7.3 (C) Risk bound

Let $\delta = \delta_*$ at the end; for now $\delta > 0$ general. Decompose, with the comparison point $m := \Pi_\delta f_0$, the bias $e = m - f_0$, and the sieve error $h := \widehat{f} - m \in \mathcal{F}_\delta - \mathcal{F}_\delta$,

$$\widehat{f} - f_0 = h + e.$$

Bias. By (14), $\|e\|_2^2 \lesssim \delta^{2\alpha}$.

Linearize about the comparison point (multiplicative r). Rather than linearize about $\mathbf{1}$ (which produces an *additive* remainder $\sim r\|h\|_2$ that cannot be absorbed at fixed r), we linearize about $m = \Pi_\delta f_0$. Note $1 - 2r \le m \le 1 + 2r$ and $\|\nabla u_m\|_\infty \lesssim 1$ ((T6), since $m \in \mathcal{F}(\alpha, 2r) \subset C^\alpha$). Write $u_{\widehat{f}} - u_m = T_m h + \widetilde{\rho}$, where T_m is the linearized forward map at m and $\widetilde{\rho}$ is its quadratic remainder. The same energy argument as Lemma 4.7, with ∇u_m in place of ∇u_0, gives $\|\widetilde{\rho}\|_2 \lesssim \|h\|_\infty\|h\|_2$. The linearization about $\mathbf{1}$, $T = T_{\mathbf{1}}$, differs from T_m by an operator-difference reflecting the r-closeness of m

to $\mathbf{1}$; the essential point is that this enters *multiplicatively*: for wavelet-packet h on the sieve at scale δ the h-dependent part of $T_m - T$ satisfies

$$T_m h - Th = -\Delta^{-1}\operatorname{div}\big(h\nabla(u_m - u_0)\big), \qquad \|T_m h - Th\|_2 \;\lesssim\; r\,\delta\,\|h\|_2, \qquad m - \mathbf{1} = O(r) \text{ in } C^\alpha, \quad (18)$$

To see the identity in (18), subtract the linearized equations: $\operatorname{div}(\nabla T_m h) = -\operatorname{div}(h\nabla u_m)$ and $\operatorname{div}(\nabla Th) = -\operatorname{div}(h\nabla u_0)$ give $\operatorname{div}\nabla(T_m h - Th) = -\operatorname{div}\big(h\nabla(u_m - u_0)\big)$, i.e. $T_m h - Th = -\Delta^{-1}\operatorname{div}\big(h\nabla(u_m - u_0)\big)$ (this is a first-order expansion of the *difference* $(T_m - T)h$, the h-independent piece $(T_m - T)0 = u_m - u_0$ being exactly accounted for by $u_m - u_0 = \Delta^{-1}\operatorname{div}((m-\mathbf{1})\nabla u_0)$). For the bound, the only subtlety is the boundary layer of $m - \mathbf{1}$: by the interior Schauder estimate $\|u_m - u_0\|_{C^{2,\beta}(K)} \le C_K\|m - \mathbf{1}\|_{C^\alpha}$ on compact $K \Subset D$ (the right side of $\operatorname{div}(m\nabla(u_m - u_0)) = \operatorname{div}((m-\mathbf{1})\nabla u_0)$ is the divergence of the vector field $(m-\mathbf{1})x/d$, whose $C^{\alpha-1}(K)$ norm is $\lesssim_K r$; (T6) applies) and the packet h is supported in $\{|x| \le \frac{3}{4}\} =: K$, so $\|h\nabla(u_m - u_0)\|_2 \le C_K r\|h\|_2$. Repeating the single-scale Fourier computation of Lemma 5.3 with the vector source $h\nabla(u_m - u_0)$ in place of xg — for the *upper* bound only, where the estimate reduces to $\|-\Delta^{-1}\operatorname{div} F\|_2 \lesssim \delta\|F\|_2$ for any vector field F whose spatial scale is δ (Lemma 5.3's upper bound (10) applied componentwise; the slow factor $\nabla(u_m - u_0)$ is $O(r)$ in L^∞ and $C^{\alpha-1}$ and does not change the scale) — yields

$$\|T_m h - Th\|_2 \;\lesssim\; \delta\, C_r\,\|h\|_2 \;\lesssim\; r\,\delta\,\|h\|_2$$

(this is the multiplicative r: the estimate scales linearly with $\|m - \mathbf{1}\|_{C^\alpha} \lesssim r$, and fails for $r = 1$). The remaining piece $(u_m - u_0)$ of $(T_m - T)$ is *independent of* h and is grouped, below, with the bias/discrepancy term (it acts as an additional $O(r)$ offset inside $u_m - u_{f_0}$; its contribution to the risk is handled with the bias in the accounting display below). Consequently the signal satisfies the recoverable lower bound

$$\|P_\delta T_m h\|_2 \;\ge\; \|P_\delta Th\|_2 - \|(T_m - T)h\|_2 \;\ge\; \big(c - Cr\big)\delta\|h\|_2 \;\ge\; \tfrac{c}{2}\,\delta\|h\|_2 \qquad \text{for } r \le r_0 := \tfrac{c}{2C}, \quad (19)$$

by Lemma 7.2 applied to the annulus packet Th and (18). Since the sieve perturbations h are by construction supported in the bulk annulus (no origin-near modes are estimated; Lemma 7.2's lower bound applies to the *whole* h), this is the recovery lower bound used in (22); on the origin content absorbed into Π_δ see Remark 7.4.

Absorption via per-scale Nikolskii on the compact sieve. On the compact sieve $\mathcal{F}_\delta \subset \{\|\cdot\|_{C^\alpha} \le 2r\}$, a difference $h = \sum_j h_j$ of two sieve elements has $\|h_j\|_{C^\alpha} \le 4r$ at each scale $2^{-j} \ge \delta$. The per-scale Nikolskii inequality and the C^α class cap give $\|h_j\|_\infty \lesssim \min\big(2^{jd/2}\|h_j\|_2,\, 2^{-j\alpha}r\big)$. Summing over the $\lesssim \log(1/\delta)$ scales $2^{-j} \ge \delta$, and using Lemma 4.7's $\|\widetilde{\rho}\|_2 \lesssim \|h\|_\infty\|h\|_2$ together with the pointwise triangle inequality $\|h\|_\infty \le \sum_j \|h_j\|_\infty$ (the pieces are summable in L^∞ since $\|h_j\|_\infty \lesssim 2^{-j\alpha}r$ and $\alpha > 0$),

$$\frac{\|\widetilde{\rho}\|_2}{\delta\|h\|_2} \;\le\; \frac{\|h\|_\infty}{\delta} \;\le\; \delta^{-1}\sum_{2^{-j}\ge\delta}\|h_j\|_\infty \;\lesssim\; \delta^{-1}\sum_{2^{-j}\ge\delta}\min\big(2^{jd/2}\|h_j\|_2,\, 2^{-j\alpha}r\big).$$

Using the bootstrap $\|h\|_2 \le 2r$ and the uniform C^α bound, each summand is $\lesssim 2^{-j\alpha}r$, so $\delta^{-1}\sum_j 2^{-j\alpha}r \lesssim \delta^{-1}r\delta^\alpha = r\delta^{\alpha-1} \to 0$ *because* $\alpha > 1$. Hence, at the scale of interest and quantitatively (no undefined exponent),

$$\|\widetilde{\rho}\|_2 \;\le\; r\,\delta^{\alpha-1}\big(\delta\|h\|_2\big) \;=:\; \omega_\delta\,\delta\|h\|_2, \qquad \omega_\delta := r\delta^{\alpha-1} \to 0 \;\; (\alpha > 1). \quad (20)$$

This is the genuine multiplicative absorption, valid for all $\alpha > 1$, all $d \ge 2$.

Basic inequality and risk. Write, abusing notation, $\Upsilon(g) = \|P_\delta u_g - Y_\delta\|_2^2$, $\Delta Y := P_\delta(u_{f_0} - Y_\delta) = n^{-1/2}\xi_\delta$ (the projected negative noise), and set

$$X \;:=\; P_\delta\big(u_{\widehat{f}} - u_m\big), \qquad Z \;:=\; P_\delta\big(u_{f_0} - u_m\big).$$

From $\Upsilon(\widehat{f}) \le \Upsilon(m)$, expanding the squares as $\|X-(\Delta Y+Z)\|_2^2 \le \|\Delta Y+Z\|_2^2$ and cancelling the common $\|\Delta Y+Z\|_2^2$,

$$\|X\|_2^2 \;\le\; 2\,\big|\langle X, \Delta Y+Z\rangle\big| \;\le\; 2\|X\|_2\big(\|\Delta Y\|_2+\|Z\|_2\big),$$

so (dividing by $\|X\|_2$; if $X=0$ the conclusion is trivial)

$$\|X\|_2 \;\le\; 2\,\|\Delta Y\|_2 + 2\,\|Z\|_2 \;=\; 2n^{-1/2}\|\xi_\delta\|_2 + 2\|P_\delta(u_{f_0}-u_m)\|_2. \tag{21}$$

On the left, $u_{\widehat{f}}-u_m = T_m h+\widetilde{\rho}$ with $\|\widetilde{\rho}\|_2 \le \omega_\delta\,\delta\|h\|_2$ by (20), hence by (19)

$$\|X\|_2 \;\ge\; \|P_\delta T_m h\|_2 - \|\widetilde{\rho}\|_2 \;\ge\; \tfrac{c}{2}\delta\|h\|_2 - \omega_\delta\delta\|h\|_2 \;\ge\; \tfrac{c}{4}\,\delta\|h\|_2 \qquad (\delta\le\delta_0).$$

For the right side of (21), the *discrepancy* $u_{f_0}-u_m$ is the solution error generated by the approximation residual $f_0-m = f_0-\Pi_\delta f_0 = -e$; by the fine-scale bias bound (next paragraph), $\|u_{f_0}-u_m\|_2 \lesssim \delta^{1+\alpha}$, i.e. $\|Z\|_2 = \|P_\delta(u_{f_0}-u_m)\|_2 \lesssim \delta^{1+\alpha}$. Combining the last two displays with (21),

$$\tfrac{c}{4}\delta\|h\|_2 \;\lesssim\; n^{-1/2}\|\xi_\delta\|_2 + \delta^{1+\alpha} + \omega_\delta\,\delta\|h\|_2, \tag{22}$$

and the $\omega_\delta\delta\|h\|_2$ term is absorbed on the left for $\delta\le\delta_0$ (since $\omega_\delta\to 0$).

Fine-scale bias. Write $w := u_{f_0}-u_m$; both u_{f_0} and u_m vanish on ∂D (the Dirichlet condition of the very-weak formulation (4)), so $w\in H_0^1$ solves $\operatorname{div}(f_0\nabla w) = \operatorname{div} G$, $G := e\nabla u_m$, with $\|G\|_2 \le \|e\|_2\|\nabla u_m\|_\infty \lesssim \delta^\alpha$ by (14) and $\|\nabla u_m\|_\infty \lesssim 1$ (T6). Split $G = G_{\mathrm{int}}+G_{\mathrm{col}}$, $G_{\mathrm{col}} = G\,\mathbf{1}_S$ the part in a collar $S=\{|x|>1-s\}$ of width s around ∂D. Two points: (i) *collar remainder*: by Hölder and (14), $\|e\|_{L^2(S)} \le \|e\|_{L^p(S)}|S|^{\frac12-\frac1p} \lesssim s^{\frac12-\frac1p}\delta^\alpha$, with $p>2$ taken large (the C^α-cap tail is Nikolskii-small, $\|e\|_{L^p(S)} \lesssim \delta^\alpha$, the projected region $O(r)$); for $s\lesssim\delta$ the collar is $\lesssim \delta^{\alpha+\frac14} = o(\delta^\alpha)$, and no $r\delta^\alpha$ term appears — collar content is projected into Π_δ; (ii) *interior gain*: each scale-δ bump of e, at distance $\gtrsim s$ from ∂D, is mean-zero, so by the exact Fourier mechanism of Lemma 5.3 (the $|\xi|^{-1}$-gain on the loaded shell; the harmonic correction (11) applies since $\operatorname{supp} G_{\mathrm{int}}$ is interior) $\|G_{\mathrm{int}}\|_{H^{-1}} \lesssim \delta\|e\|_2 \lesssim \delta^{\alpha+1}$. Combining, testing against $\phi = -\Delta_{f_0}^{-1}\psi$ as in Lemma 4.6 and using Poincaré on $w\in H_0^1$,

$$\|w\|_2 = \|u_{f_0}-u_m\|_2 \;\lesssim\; \|G\|_{H^{-1}} \;\lesssim\; \delta\|e\|_2 + \delta^{\alpha+\frac14} \;\lesssim\; \delta^{1+\alpha}. \tag{23}$$

Thus $\|Z\|_2 = \|P_\delta(u_{f_0}-u_m)\|_2 \lesssim \delta^{1+\alpha}$.

Noise enters linearly (no union bound necessary). In the comparison inequality the noise appears only through the *linear* functional $h\mapsto n^{-1/2}\langle\xi_\delta, P_\delta T h\rangle$ evaluated at the random point $h=\widehat{f}-\Pi_\delta f_0$; by Cauchy–Schwarz, $n^{-1/2}|\langle\xi_\delta,P_\delta Th\rangle| \le n^{-1/2}\|\xi_\delta\|_2\,\|P_\delta Th\|_2$, and after division by $\|P_\delta Th\|_2$ this contributes $n^{-1/2}\|\xi_\delta\|_2$ on the event $\{\widehat{f}\in\mathcal{F}_\delta\}$, uniformly. Because the projected model is finite-dimensional and the functional is linear, $\mathbb{E}\|\xi_\delta\|_2^2 = \dim V_\delta \asymp \delta^{-d}$, and *no supremum / union bound over the sieve is needed here*: a logarithmic factor would arise only if one insisted on a *uniform*-in-g comparison, which the least-squares basic inequality avoids. Account for every term of (22): after absorbing $\omega_\delta\delta\|h\|_2$ and dividing by δ, $\|h\|_2 \lesssim \delta^{-1}\big(n^{-1/2}\|\xi_\delta\|_2+\delta^{1+\alpha}\big)$, so squaring (and using $\mathbb{E}\|\xi_\delta\|_2^2 \asymp \delta^{-d}$) $\mathbb{E}\|h\|_2^2 \lesssim \delta^{-d-2}/n + \delta^{2\alpha}$: the discrepancy $\delta^{1+\alpha}$ contributes $\delta^{2\alpha}$, not $r^2\delta^{2\alpha-2}$ — the boundary collar/collar content responsible for the latter has been projected into the bias, whose u-image uses the interior degree-one gain (23). With the balance $\delta=\delta_* = n^{-1/(2\alpha+d+2)}$ chosen so $\delta_*^{-d-2}/n = \delta_*^{2\alpha}$, one therefore obtains the *log-free* estimate

$$\mathbb{E}\|\widehat{f}-f_0\|_2^2 \lesssim \|e\|_2^2 + \mathbb{E}\|h\|_2^2 \lesssim \delta_*^{2\alpha} + \frac{\delta_*^{-d-2}}{n} \asymp n^{-\frac{2\alpha}{2\alpha+d+2}}.$$

All contributing terms (bias, discrepancy, noise) are $\lesssim \delta_*^{2\alpha}$ at the balance; the upper bound is free of logarithmic loss.

8 Conclusion and consistency checks

- **Rate.** Lower bound (Proposition 6.1): $n^{-2\alpha/(2\alpha+d+2)}$ (no log). Upper bound (Proposition 7.1): $n^{-2\alpha/(2\alpha+d+2)}$ (no log: the noise enters the least-squares basic inequality only through a *linear* functional, so no union bound is needed). The two agree, so the minimax rate is $n^{-2\alpha/(2\alpha+d+2)}$ up to logarithmic factors.
- **Interpretation.** The exponent denominator $2\alpha + d + 2$ exceeds the direct-problem $2\alpha + d$ by exactly 2, reflecting *one* degree of smoothing of the linearized inverse T (order -1), a consequence of the transport structure $\Delta T = I + \frac{x\cdot\nabla}{d}$ and the L^2-invertibility of $I + \frac{x\cdot\nabla}{d}$ on $L^2(D)$ (Lemmas 4.1, 4.3 and the exact Fourier identity Lemma 5.1).
- **Role of $\alpha > 1$.** Used decisively in both bounds to make the quadratic remainder $R(h)$ (Lemma 4.7) negligible against the linear signal Th at the minimax scale (ratio $\asymp \delta_*^{\alpha-1} \to 0$), cf. Remark 4.8. For $\alpha \leq 1$ the linearization is not faithful at scale δ_*; the rate would then require a genuinely nonlinear analysis (we make no claim in that regime).
- **Role of small r.** $r \leq r_0(\alpha, d)$ is used for (i) uniform ellipticity $1 - r \leq f \leq 1 + r$, (ii) absorbing the quadratic remainder into the linear signal, and (iii) the uniform apriori bound $\|u_f\|_\infty \lesssim 1$. All structural constants and r_0 depend only on (α, d, ψ); C is uniform in $r \in (0, r_0]$ and n; the lower-bound c carries the r^2 factor of the shrinking class.
- **Model handling.** u_f is defined by the very-weak formulation (4); no H^1 membership is asserted (Section 4.5). The white noise is only ever projected onto the finite-dimensional sieve V_δ, where it is a well-defined Gaussian vector, so the estimator/risk are rigorous on the white-noise model.